NPLify · P0 Technical Baseline

Deliverable 1 — Entity-Relationship Diagram & Data Model

New XP Technologies Limited

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Conventions

Convention: all monetary amounts are stored as exact decimals; every table carries `created_at` / `created_by`; audit, approvals and locking are modelled once (domain 4) and apply to all master data. Derived figures (client balance, group entitlement, offsets, custody) are **views over postings and deals — never stored**.

Seven diagrams across four domains, one schema: 70 tables and 12 views. In each diagram, entities defined in another diagram appear with no attributes and name that diagram. Relationship lines run from the referenced row to the row that stores the foreign key; `||--o{` is composition (the parent owns the child), `||..o{` a reference or a polymorphic link resolved by a type column plus an id. The deal lifecycle has three diagrams: 3a is the deal and its collection, 3b the conversion and its exceptions, 3c the settlement. Appendix A reconciles the model with the FRS; Appendix B is the data dictionary.

1 · Configuration domain

A **Client** is the commercial relationship NPL contracts with; a **Project** is one arrangement with it and owns its configuration. Partners are global counterparties; how a partner is used *in this project* is the **partner configuration** block — fees, vehicles, endpoints, destination approvals and rebate terms hang off it. Counterparty receivers are organised in **receiver groups**, the unit entitlement is tracked on, and each receiving entity holds one or more **accounts** with a currency and rail.

Diagram 1a. `PARTNER_CONFIG`, `PARTNER_PAIR`, `DESTINATION_APPROVAL`, `FEE_STRUCTURE` are defined in Diagram 1b.

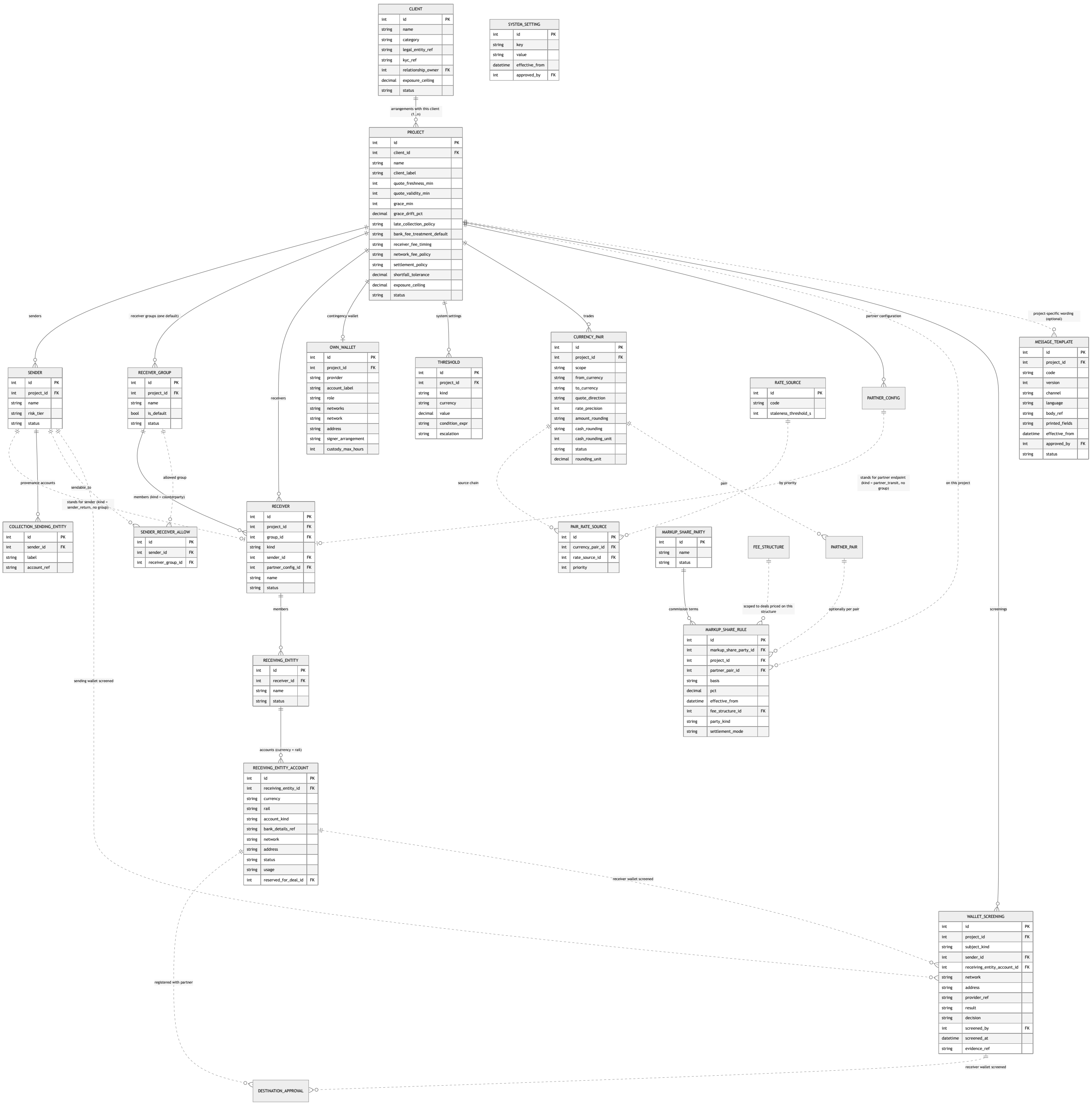
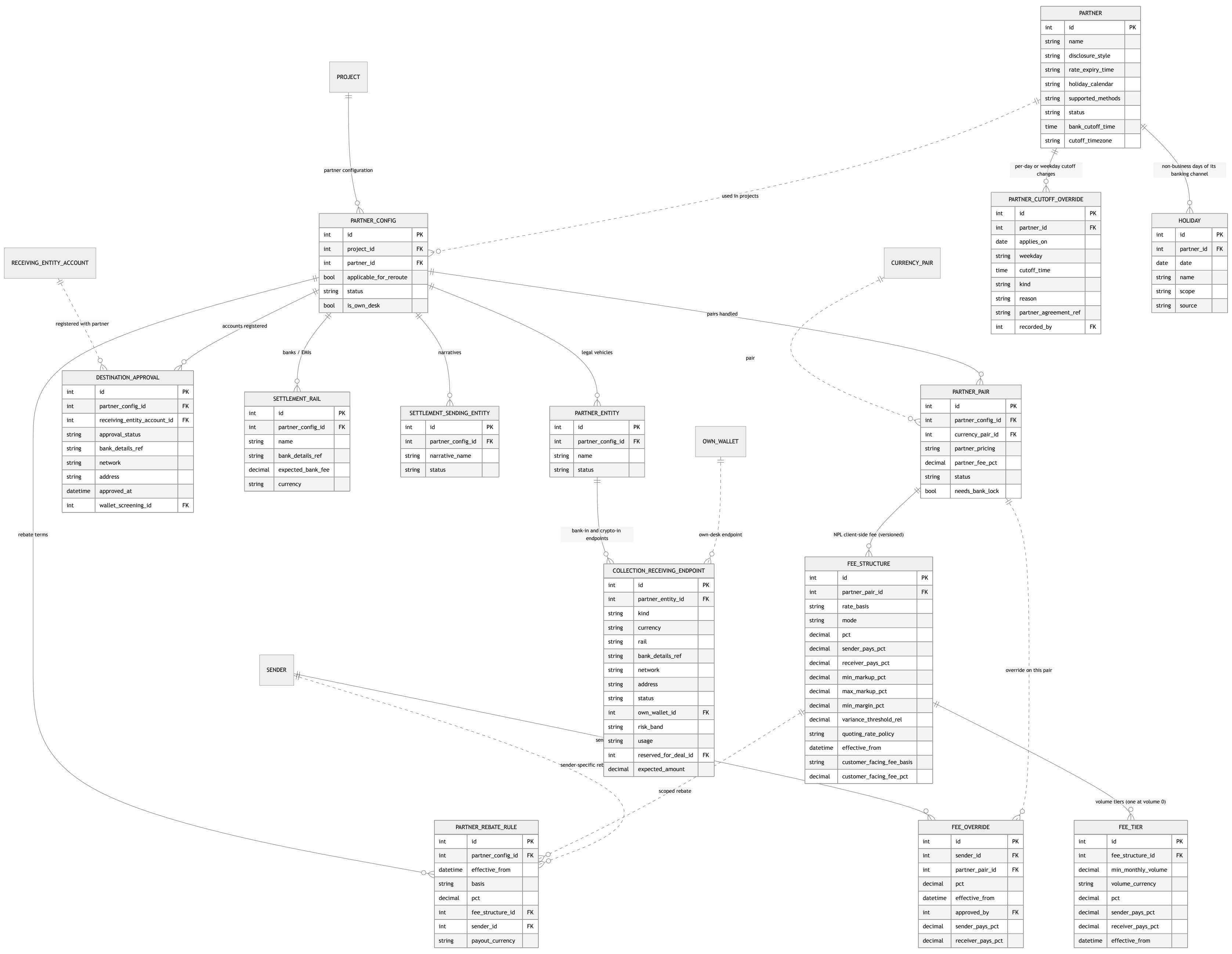

CLIENT
int id PK
string name
string category
string legal_entity_ref
string kyc_ref
int relationship_owner FK
decimal exposure_ceiling
string status
SYSTEM_SETTING
int id PK
string key
string value
datetime effective_from
int approved_by FK
arrangements with this client (1..n)
PROJECT
int id PK
int client_id FK
string name
string client_label
int quote_freshness_min
int quote_validity_min
int grace_min
decimal grace_drift_pct
string late_collection_policy
string bank_fee_treatment_default
string receiver_fee_timing
string network_fee_policy
string settlement_policy
decimal shortfall_tolerance
decimal exposure_ceiling
string status
senders
receiver groups (one default)
contingency wallet
system settings
trades
partner configuration
project-specific wording (optional)
SENDER
int id PK
int project_id FK
string name
string risk_tier
string status
RECEIVER_GROUP
int id PK
int project_id FK
string name
bool is_default
string status
OWN_WALLET
int id PK
int project_id FK
string provider
string account_label
string role
string networks
string network
string address
string signer_arrangement
int custody_max_hours
THRESHOLD
int id PK
int project_id FK
string kind
string currency
decimal value
string condition_expr
string escalation
CURRENCY_PAIR
int id PK
int project_id FK
string scope
string from_currency
string to_currency
string quote_direction
int rate_precision
string amount_rounding
string cash_rounding
int cash_rounding_unit
string status
decimal rounding_unit
RATE_SOURCE
int id PK
string code
int staleness_threshold_s
PARTNER_CONFIG
MESSAGE_TEMPLATE
int id PK
int project_id FK
string code
int version
string channel
string language
string body_ref
string printed_fields
datetime effective_from
int approved_by FK
string status
receivers
provenance accounts
sendable_to
stands for sender (kind = sender_return, no group)
members (kind = counterparty)
allowed group
source chain
pair
by priority
stands for partner endpoint (kind = partner_transit, no group)
on this project
COLLECTION_SENDING_ENTITY
int id PK
int sender_id FK
string label
string account_ref
SENDER_RECEIVER_ALLOW
int id PK
int sender_id FK
int receiver_group_id FK
RECEIVER
int id PK
int project_id FK
int group_id FK
string kind
int sender_id FK
int partner_config_id FK
string name
string status
PAIR_RATE_SOURCE
int id PK
int currency_pair_id FK
int rate_source_id FK
int priority
MARKUP_SHARE_PARTY
int id PK
string name
string status
FEE_STRUCTURE
PARTNER_PAIR
members
commission terms
scoped to deals priced on this structure
optionally per pair
screenings
MARKUP_SHARE_RULE
int id PK
int markup_share_party_id FK
int project_id FK
int partner_pair_id FK
string basis
decimal pct
datetime effective_from
int fee_structure_id FK
string party_kind
string settlement_mode
sending wallet screened
RECEIVING_ENTITY
int id PK
int receiver_id FK
string name
string status
accounts (currency × rail)
RECEIVING_ENTITY_ACCOUNT
int id PK
int receiving_entity_id FK
string currency
string rail
string account_kind
string bank_details_ref
string network
string address
string status
string usage
int reserved_for_deal_id FK
receiver wallet screened
WALLET_SCREENING
int id PK
int project_id FK
string subject_kind
int sender_id FK
int receiving_entity_account_id FK
string network
string address
string provider_ref
string result
string decision
int screened_by FK
datetime screened_at
string evidence_ref
registered with partner
receiver wallet screened
DESTINATION_APPROVAL

Diagram 1a — Configuration — project, parties, receiver groups and wallet screening

Diagram 1b. `PROJECT`, `CURRENCY_PAIR`, `SENDER`, `RECEIVING_ENTITY_ACCOUNT`, `OWN_WALLET` are defined in Diagram 1a.


PROJECT
partner configuration
used in projects
per-day or weekday cutoff changes
non-business days of its banking channel
RECEIVING_ENTITY_ACCOUNT
CURRENCY_PAIR
registered with partner
accounts registered
banks / EMIs
narratives
legal vehicles
pairs handled
pair
OWN_WALLET
rebate terms
bank-in and crypto-in endpoints
own-desk endpoint
NPL client-side fee (versioned)
SENDER
override on this pair
sen
sender-specific reb
scoped rebate
volume tiers (one at volume 0)

PARTNER

int	id	PK
string	name	
string	disclosure_style	
string	rate_expiry_time	
string	holiday_calendar	
string	supported_methods	
string	status	
time	bank_cutoff_time	
string	cutoff_timezone	

PARTNER_CUTOFF_OVERRIDE

int	id	PK
int	partner_id	FK
date	applies_on	
string	weekday	
time	cutoff_time	
string	kind	
string	reason	
string	partner_agreement_ref	
int	recorded_by	FK

HOLIDAY

int	id	PK
int	partner_id	FK
date	date	
string	name	
string	scope	
string	source	

PARTNER_CONFIG

int	id	PK
int	project_id	FK
int	partner_id	FK
bool	applicable_for_reroute	
string	status	
bool	is_own_desk	

DESTINATION_APPROVAL

int	id	PK
int	partner_config_id	FK
int	receiving_entity_account_id	FK
string	approval_status	
string	bank_details_ref	
string	network	
string	address	
datetime	approved_at	
int	wallet_screening_id	FK

SETTLEMENT_RAIL

int	id	PK
int	partner_config_id	FK
string	name	
string	bank_details_ref	
decimal	expected_bank_fee	
string	currency	

SETTLEMENT_SENDING_ENTITY

int	id	PK
int	partner_config_id	FK
string	narrative_name	
string	status	

PARTNER_ENTITY

int	id	PK
int	partner_config_id	FK
string	name	
string	status	

PARTNER_PAIR

int	id	PK
int	partner_config_id	FK
int	currency_pair_id	FK
string	partner_pricing	
decimal	partner_fee_pct	
string	status	
bool	needs_bank_lock	

COLLECTION_RECEIVING_ENDPOINT

int	id	PK
int	partner_entity_id	FK
string	kind	
string	currency	
string	rail	
string	bank_details_ref	
string	network	
string	address	
string	status	
int	own_wallet_id	FK
string	risk_band	
string	usage	
int	reserved_for_deal_id	FK
decimal	expected_amount	

FEE_STRUCTURE

int	id	PK
int	partner_pair_id	FK
string	rate_basis	
string	mode	
decimal	pct	
decimal	sender_pays_pct	
decimal	receiver_pays_pct	
decimal	min_markup_pct	
decimal	max_markup_pct	
decimal	min_margin_pct	
decimal	variance_threshold_rel	
string	quoting_rate_policy	
datetime	effective_from	
string	customer_facing_fee_basis	
decimal	customer_facing_fee_pct	

PARTNER_REBATE_RULE

int	id	PK
int	partner_config_id	FK
datetime	effective_from	
string	basis	
decimal	pct	
int	fee_structure_id	FK
int	sender_id	FK
string	payout_currency	

FEE_OVERRIDE

int	id	PK
int	sender_id	FK
int	partner_pair_id	FK
decimal	pct	
datetime	effective_from	
int	approved_by	FK
decimal	sender_pays_pct	
decimal	receiver_pays_pct	

FEE_TIER

int	id	PK
int	fee_structure_id	FK
decimal	min_monthly_volume	
string	volume_currency	
decimal	pct	
decimal	sender_pays_pct	
decimal	receiver_pays_pct	
datetime	effective_from	

Diagram 1b — Configuration — partner configuration block, fees and tiers

Notes

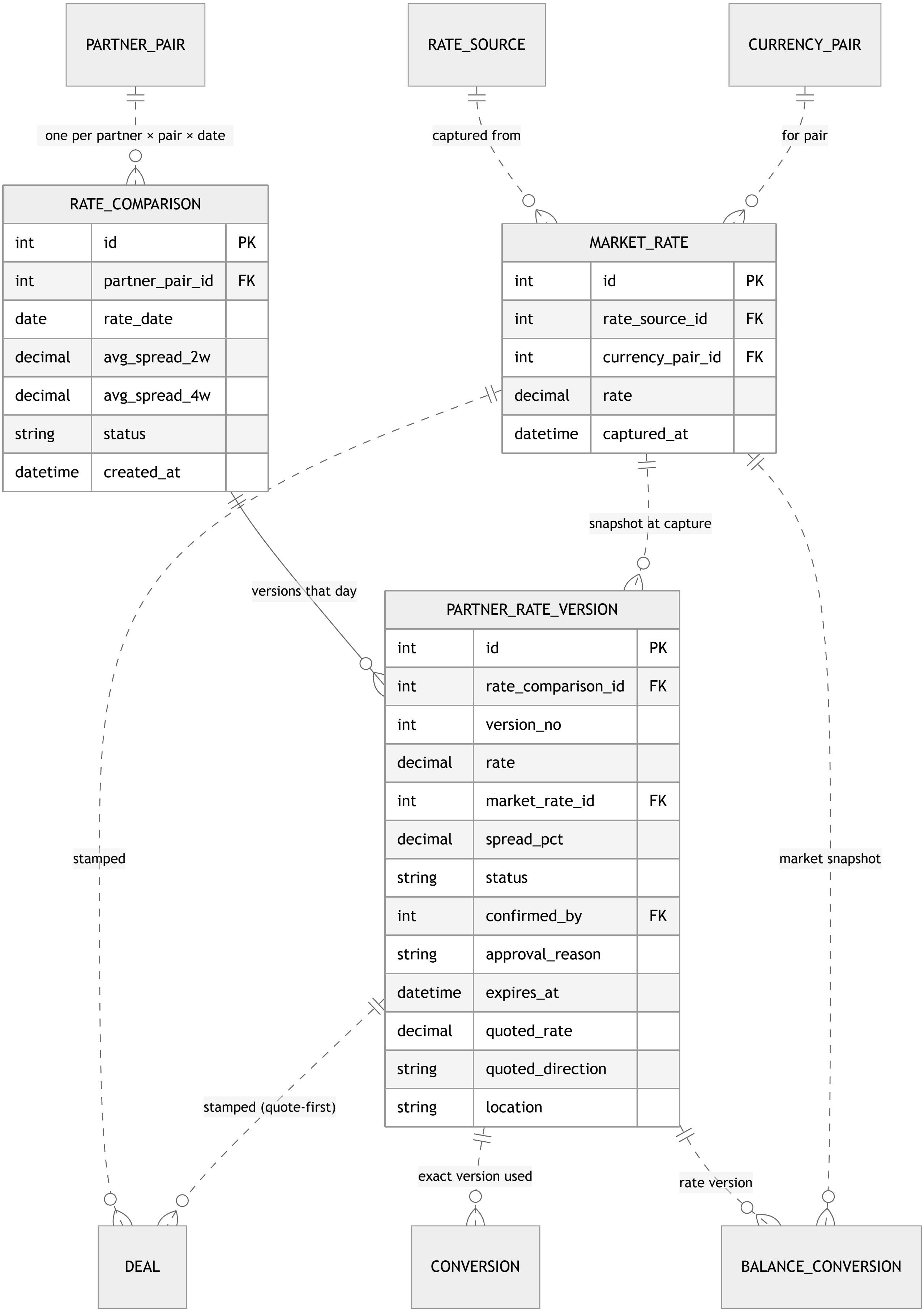
- A **Client** is the commercial relationship NPL contracts with; a **Project** is one arrangement with it and owns its configuration. Counterparty receivers sit in **receiver groups**: every project has one default group and every counterparty receiver belongs to exactly one group. Entitlement is tracked per group, so payouts within a group are ordinary settlement and only payouts across groups are offsets; a project with a single group never has an offset.
- A **Receiver** has a kind: `counterparty` for a client receiver; `sender_return` stands for a sender so refunds use the ordinary payout path and the partner's destination approval; `partner_transit` stands for a partner's collection endpoint so the hop of a two-leg route is an ordinary payout. `sender_return` and `partner_transit` receivers belong to no group (`group_id` is null). They never appear on forward deals or allow lists — both validate kind = counterparty — and a payout to them is attributed to the group of the leg that pays them, so no entitlement moves between groups on a refund or a hop. Groups exist for entitlement and offset accounting; refund exclusion is a receiver-kind rule, not a group.
- A receiving entity holds one or more **accounts**, each with a currency and rail (EUR via SWIFT and EUR via SEPA are two accounts). An account is a bank account or a wallet (`account_kind`): a bank account carries `bank_details_ref` and a bank rail; a wallet carries `network` and `address` explicitly, mirroring the partner-side `COLLECTION_RECEIVING_ENDPOINT`, so addresses can be checksum-validated, checked against the rail, and found when the same address is registered with two partners. Destination approvals and settlement lines point at the account. No table stores actual bank details: `bank_details_ref` points at the external store.
- Fees are per partner × pair (`FEE_STRUCTURE` under `PARTNER_PAIR`), so choosing the partner is part of pricing. `PARTNER.disclosure_style` records whether the partner's markup is visible; `partner_fee_pct` is known only for fee_on_market partners.
- A partner operates through one or more legal vehicles (`PARTNER_ENTITY`: PT-tour, PT Global Inc); the deal stamps the vehicle at quote so paperwork names the right company. Each vehicle owns the endpoints a sender can pay into (`COLLECTION_RECEIVING_ENDPOINT`): kind = bank with a currency and `bank_details_ref`, or kind = wallet with `network` and `address`. The partner side therefore mirrors the client side — `PARTNER_CONFIG` → `PARTNER_ENTITY` → `COLLECTION_RECEIVING_ENDPOINT` as `RECEIVER` → `RECEIVING_ENTITY` → `RECEIVING_ENTITY_ACCOUNT`. An endpoint also carries a nullable `rail`: a bank endpoint must have it before it becomes active, and a wallet may leave it null but must match `network` when it is set; the endpoint's kind must match the leg's `collection_method`. Cash-in has no endpoint record; the leg carries the city and the collection the token. The collected cash amount is rounded by the pair's `cash_rounding` option — truncate, nearest or round up — to `cash_rounding_unit` (100 / 500 / 1,000 by currency), the same menu as payout rounding.
- Destination approval is per partner: `DESTINATION_APPROVAL` is an account of the project's receiving entity as registered with this partner, carrying the partner's approval status and the payout details the partner holds. Those details follow the account's kind: `bank_details_ref` for a bank account, `network` and `address` for a wallet. `SETTLEMENT_SENDING_ENTITY` is the narrative on an outgoing transfer; `SETTLEMENT_RAIL` is the bank/EMI that executes it; neither holds money. `COLLECTION_SENDING_ENTITY` is provenance only and never a payout destination.
- `OWN_WALLET` has a role: reroute contingency (AQN-LT-Sub at Aquanow), own desk, or a collection bridge on a network a partner lacks, forwarded to the partner as a hop within the wallet's custody limit; `PARTNER_CONFIG.applicable_for_reroute` defines the alternate-partner set. `PARTNER.supported_methods` declares what a partner offers — cash and local deposits have no standing endpoint to read it from. `WALLET_REGISTRY` lists every known address with its owner and role. Under an **own-desk** partner configuration (`is_own_desk`, the NPL-GR game-reseller project) NPL itself is the

converting party: its pairs are priced market plus NPL's markup and its collection endpoints are NPL's own wallets (`COLLECTION_RECEIVING_ENDPOINT.own_wallet_id`); the custody view reports every such balance.

- **Fees.** A `FEE_OVERRIDE` may carry its own sender / receiver split instead of inheriting the structure's proportion. `FEE_TIER` rows under a structure price by the project's calendar-month volume; a leg that takes the month past a threshold is split across two tiers. `customer_facing_fee_basis` / `customer_facing_fee_pct` on the structure say what the customer's quote prints when it differs from the recorded parts; the printed "source rate" is derived at quote, never typed. `CURRENCY_PAIR.amount_rounding` is a menu — truncate, nearest or round up, to the unit or to `rounding_unit` — and a pair may have the same currency on both sides.
- **Who else earns on a deal.** `MARKUP_SHARE_PARTY` / `MARKUP_SHARE_RULE` record a share NPL pays — or, with basis `share_of_net_markup` and `party_kind` client_party, the customer who provides a sub-account and shares NPL's markup (Raeen): the base is the leg's fee parts less its partner cost, rebates excluded, several parties may hold rules with different shares, and `settlement_mode` party_retains means the party holds the converted money and pays NPL the remainder monthly. `PARTNER_REBATE_RULE` records the rebate a partner pays NPL, with a basis (percentage of amount or of the partner's fee) and an optional scope by sender or fee structure; the rebate is NPL's alone and is reported as its own stream.
- **Cutoff, screening, single use.** `PARTNER.bank_cutoff_time` (the partner's own clock, beside its rate expiry and holiday calendar), its per-day changes in `PARTNER_CUTOFF_OVERRIDE` (an agreed extension or an early close) and `PARTNER_PAIR.needs_bank_lock` drive same-day or next-day conversion and, on bank rails, same-day or next-day settlement. Cash legs carry their city (`DEAL.cash_location`) and cash rate versions theirs. `WALLET_SCREENING` records the screening of a sending or receiving wallet address, by Operations, and the endpoint's `risk_band` says which decisions may land on it. An endpoint or a receiving-entity account may be `single_use`, reserved for one deal (ad hoc local deposit accounts). `HOLIDAY` rows under the partner are the dates behind every "next business day"; `MESSAGE_TEMPLATE` is the versioned wording of outgoing messages and declares the fields it may print — a quote template prints sender-facing figures only.

2 · Rates domain

Diagram 2. `RATE_SOURCE`, `CURRENCY_PAIR` are defined in Diagram 1a; `PARTNER_PAIR` in 1b; `DEAL` in 3a; `CONVERSION`, `BALANCE_CONVERSION` in 3b.


PARTNER_PAIR
RATE_SOURCE
CURRENCY_PAIR
one per partner × pair × date
captured from
for pair
snapshot at capture
versions that day
stamped
market snapshot
stamped (quote-first)
exact version used
rate version
DEAL
CONVERSION
BALANCE_CONVERSION

RATE_COMPARISON

int	id	PK
int	partner_pair_id	FK
date	rate_date	
decimal	avg_spread_2w	
decimal	avg_spread_4w	
string	status	
datetime	created_at	

MARKET_RATE

int	id	PK
int	rate_source_id	FK
int	currency_pair_id	FK
decimal	rate	
datetime	captured_at	

PARTNER_RATE_VERSION

int	id	PK
int	rate_comparison_id	FK
int	version_no	
decimal	rate	
int	market_rate_id	FK
decimal	spread_pct	
string	status	
int	confirmed_by	FK
string	approval_reason	
datetime	expires_at	
decimal	quoted_rate	
string	quoted_direction	
string	location	

Diagram 2 — Rates

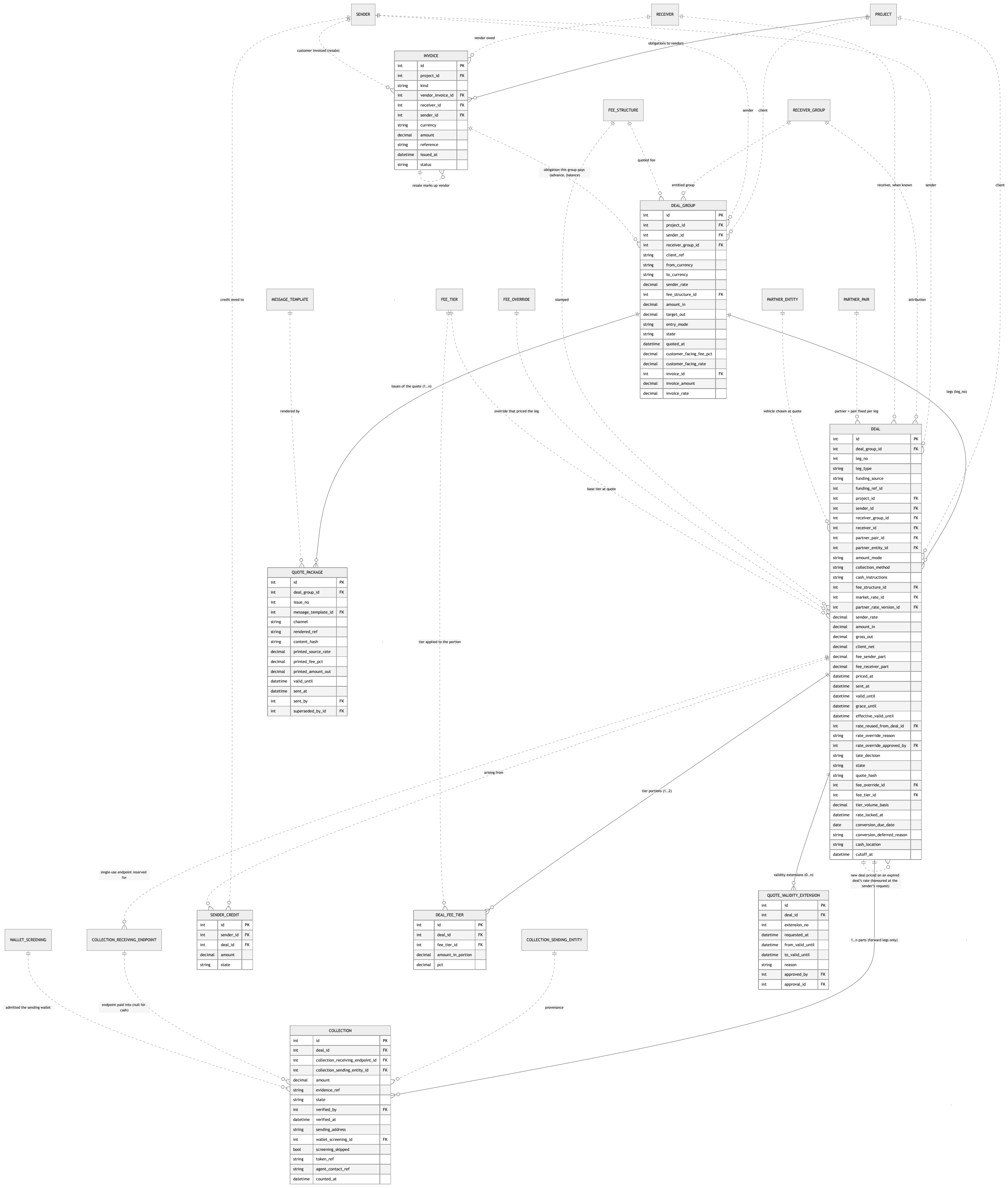
Notes

- The rate comparison for the day is created lazily by the first deal that needs it, one per partner × pair × date, and reused by every deal that day. A partner may give a new rate during the day, so one date can hold several versions. Rolling averages are built from confirmed versions only.
- A version is breached when spread > avg_2w × (1 + variance_threshold_rel); quoting and conversion on that partner × pair freeze until Finance approves with a reason, rejects, or asks for a better rate. Versions are never overwritten; every deal leg, conversion and balance conversion references its exact version.
- Cross-source slippage is a query over `MARKET_RATE` across sources for the same pair and window.
- A version stores the rate **as the partner quoted it** (`quoted_rate`, `quoted_direction`); `rate` is normalised to the pair's direction and derived — 1 ÷ quoted when the directions differ — so Ali's 1.1738 becomes 0.851934 without a hand inversion, and a direction slip is caught by the breach check. Cash pairs are priced per `location`: INR cash in Delhi and in Mumbai are two versions on the same day.

3 · Deal lifecycle domain

3a · Deal group, legs, pricing stamps, invoice, collection

Diagram 3a. `PROJECT`, `SENDER`, `RECEIVER_GROUP`, `RECEIVER`, `COLLECTION_SENDING_ENTITY`, `WALLET_SCREENING`, `MESSAGE_TEMPLATE` are defined in Diagram 1a; `PARTNER_PAIR`, `PARTNER_ENTITY`, `FEE_STRUCTURE`, `FEE_OVERRIDE`, `FEE_TIER`, `COLLECTION_RECEIVING_ENDPOINT` in 1b.


SENDER
RECEIVER
PROJECT
vendor owed
customer invoiced (resale)
obligations to vendors
resale marks up vendor
FEE_STRUCTURE
sender
client
RECEIVER_GROUP
obligation this group pays (advance, balance)
quoted fee
entitled group
receiver, when known
sender
client
credit owed to
MESSAGE_TEMPLATE
FEE_TIER
FEE_OVERRIDE
stamped
PARTNER_ENTITY
PARTNER_PAIR
attribution
issues of the quote (1..n)
legs (leg_no)
rendered by
override that priced the leg
vehicle chosen at quote
partner × pair fixed per leg
base tier at quote
tier applied to the portion
arising from
tier portions (1..2)
single-use endpoint reserved for
validity extensions (0..n)
new deal priced on an expired deal's rate (honoured at the sender's request)
WALLET_SCREENING
COLLECTION_RECEIVING_ENDPOINT
COLLECTION_SENDING_ENTITY
1..n parts (forward legs only)
admitted the sending wallet
endpoint paid into (null for cash)
provenance

INVOICE

int	id	PK
int	project_id	FK
string	kind	
int	vendor_invoice_id	FK
int	receiver_id	FK
int	sender_id	FK
string	currency	
decimal	amount	
string	reference	
datetime	issued_at	
string	status	

DEAL_GROUP

int	id	PK
int	project_id	FK
int	sender_id	FK
int	receiver_group_id	FK
string	client_ref	
string	from_currency	
string	to_currency	
decimal	sender_rate	
int	fee_structure_id	FK
decimal	amount_in	
decimal	target_out	
string	entry_mode	
string	state	
datetime	quoted_at	
decimal	customer_facing_fee_pct	
decimal	customer_facing_rate	
int	invoice_id	FK
decimal	invoice_amount	
decimal	invoice_rate	

DEAL

int	id	PK
int	deal_group_id	FK
int	leg_no	
string	leg_type	
string	funding_source	
int	funding_ref_id	
int	project_id	FK
int	sender_id	FK
int	receiver_group_id	FK
int	receiver_id	FK
int	partner_pair_id	FK
int	partner_entity_id	FK
string	amount_mode	
string	collection_method	
string	cash_instructions	
int	fee_structure_id	FK
int	market_rate_id	FK
int	partner_rate_version_id	FK
decimal	sender_rate	
decimal	amount_in	
decimal	gross_out	
decimal	client_net	
decimal	fee_sender_part	
decimal	fee_receiver_part	
datetime	priced_at	
datetime	sent_at	
datetime	valid_until	
datetime	grace_until	
datetime	effective_valid_until	
int	rate_reused_from_deal_id	FK
string	rate_override_reason	
int	rate_override_approved_by	FK
string	late_decision	
string	state	
string	quote_hash	
int	fee_override_id	FK
int	fee_tier_id	FK
decimal	tier_volume_basis	
datetime	rate_locked_at	
date	conversion_due_date	
string	conversion_deferred_reason	
string	cash_location	
datetime	cutoff_at	

QUOTE_PACKAGE

int	id	PK
int	deal_group_id	FK
int	issue_no	
int	message_template_id	FK
string	channel	
string	rendered_ref	
string	content_hash	
decimal	printed_source_rate	
decimal	printed_fee_pct	
decimal	printed_amount_out	
datetime	valid_until	
datetime	sent_at	
int	sent_by	FK
int	superseded_by_id	FK

SENDER_CREDIT

int	id	PK
int	sender_id	FK
int	deal_id	FK
decimal	amount	
string	state	

DEAL_FEE_TIER

int	id	PK
int	deal_id	FK
int	fee_tier_id	FK
decimal	amount_in_portion	
decimal	pct	

QUOTE_VALIDITY_EXTENSION

int	id	PK
int	deal_id	FK
int	extension_no	
datetime	requested_at	
datetime	from_valid_until	
datetime	to_valid_until	
string	reason	
int	approved_by	FK
int	approval_id	FK

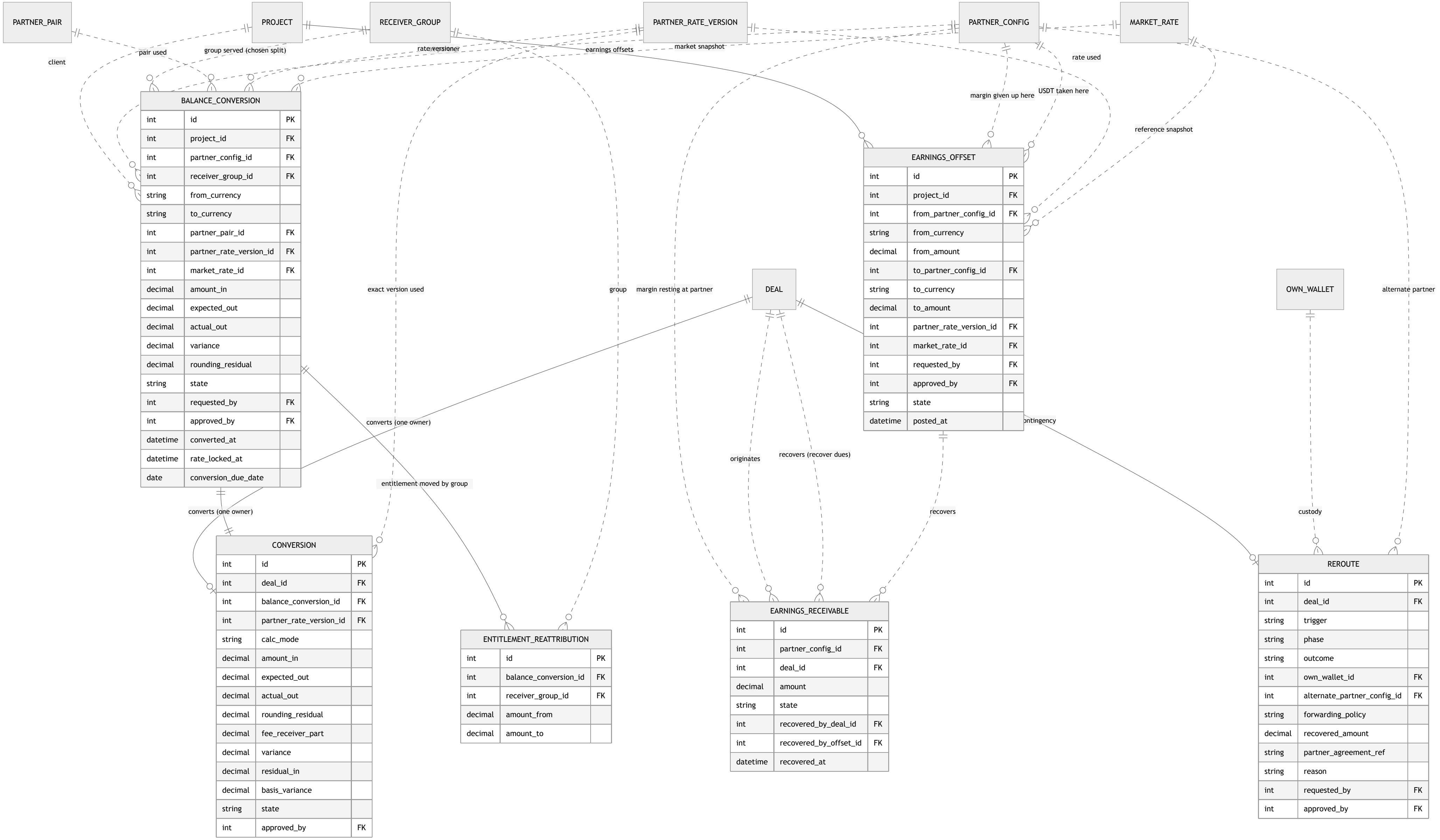
COLLECTION

int	id	PK
int	deal_id	FK
int	collection_receiving_endpoint_id	FK
int	collection_sending_entity_id	FK
decimal	amount	
string	evidence_ref	
string	state	
int	verified_by	FK
datetime	verified_at	
string	sending_address	
int	wallet_screening_id	FK
bool	screening_skipped	
string	token_ref	
string	agent_contact_ref	
datetime	counted_at	

Diagram 3a — Deal lifecycle — deal group, legs, pricing stamps, invoice, collection

3b · Conversion, reroute, earnings receivable, balance conversion, entitlement re-attribution

Diagram 3b. `PROJECT`, `RECEIVER_GROUP`, `OWN_WALLET` are defined in Diagram 1a; `PARTNER_CONFIG`, `PARTNER_PAIR` in 1b; `PARTNER_RATE_VERSION`, `MARKET_RATE` in 2; `DEAL` in 3a.


PARTNER_PAIR
PROJECT
RECEIVER_GROUP
PARTNER_RATE_VERSION
PARTNER_CONFIG
MARKET_RATE
client
pair used
group served (chosen split)
rate versioner
earnings offsets
market snapshot
rate used
margin given up here
USDT taken here
reference snapshot
exact version used
group
margin resting at partner
DEAL
OWN_WALLET
alternate partner
converts (one owner)
ontingency
originates
recovers (recover dues)
entitlement moved by group
converts (one owner)
recovers
custody

BALANCE_CONVERSION		
int	id	PK
int	project_id	FK
int	partner_config_id	FK
int	receiver_group_id	FK
string	from_currency	
string	to_currency	
int	partner_pair_id	FK
int	partner_rate_version_id	FK
int	market_rate_id	FK
decimal	amount_in	
decimal	expected_out	
decimal	actual_out	
decimal	variance	
decimal	rounding_residual	
string	state	
int	requested_by	FK
int	approved_by	FK
datetime	converted_at	
datetime	rate_locked_at	
date	conversion_due_date	

EARNINGS_OFFSET		
int	id	PK
int	project_id	FK
int	from_partner_config_id	FK
string	from_currency	
decimal	from_amount	
int	to_partner_config_id	FK
string	to_currency	
decimal	to_amount	
int	partner_rate_version_id	FK
int	market_rate_id	FK
int	requested_by	FK
int	approved_by	FK
string	state	
datetime	posted_at	

CONVERSION		
int	id	PK
int	deal_id	FK
int	balance_conversion_id	FK
int	partner_rate_version_id	FK
string	calc_mode	
decimal	amount_in	
decimal	expected_out	
decimal	actual_out	
decimal	rounding_residual	
decimal	fee_receiver_part	
decimal	variance	
decimal	residual_in	
decimal	basis_variance	
string	state	
int	approved_by	FK

ENTITLEMENT_REATTRIBUTION		
int	id	PK
int	balance_conversion_id	FK
int	receiver_group_id	FK
decimal	amount_from	
decimal	amount_to	

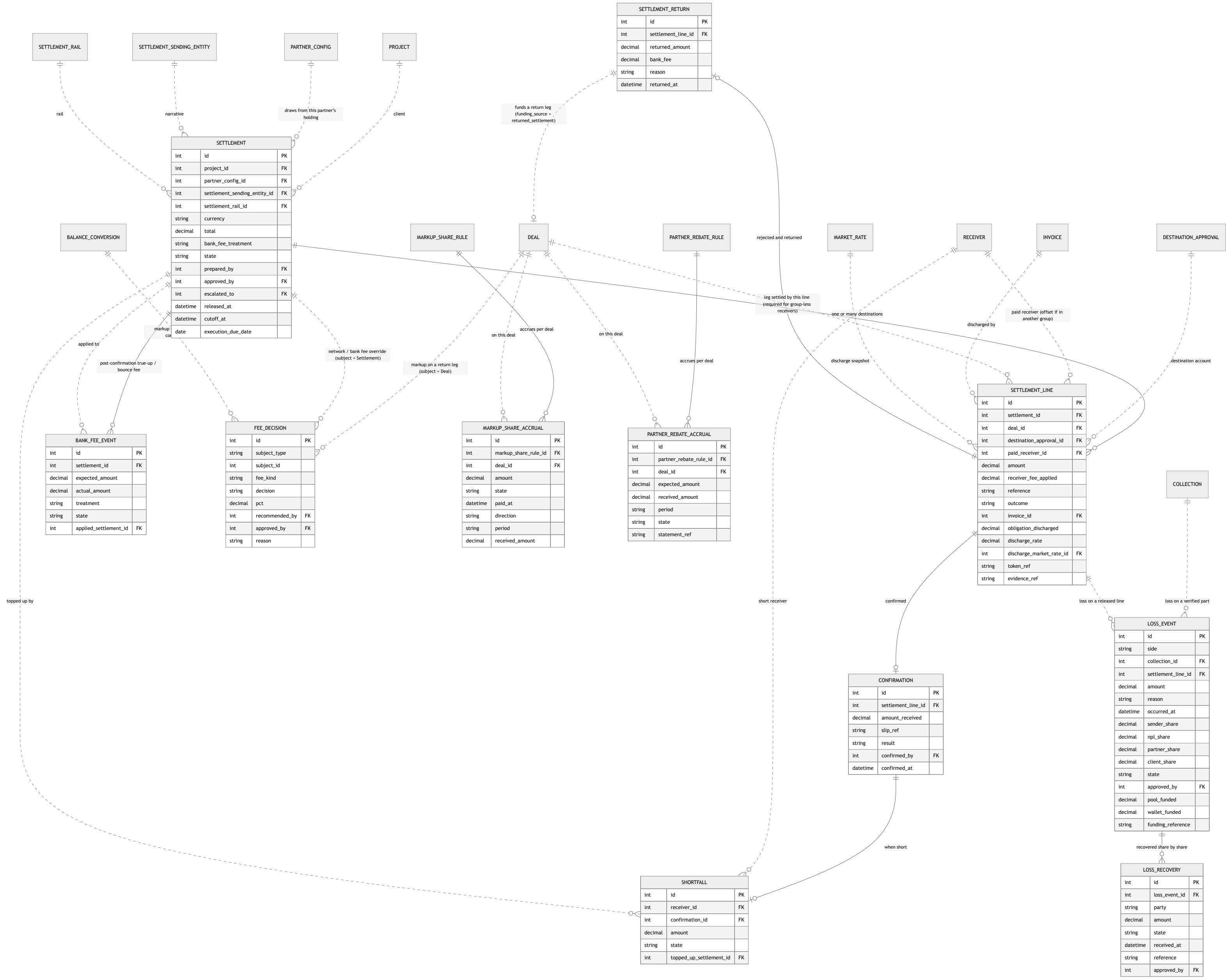
EARNINGS_RECEIVABLE		
int	id	PK
int	partner_config_id	FK
int	deal_id	FK
decimal	amount	
string	state	
int	recovered_by_deal_id	FK
int	recovered_by_offset_id	FK
datetime	recovered_at	

REROUTE		
int	id	PK
int	deal_id	FK
string	trigger	
string	phase	
string	outcome	
int	own_wallet_id	FK
int	alternate_partner_config_id	FK
string	forwarding_policy	
decimal	recovered_amount	
string	partner_agreement_ref	
string	reason	
int	requested_by	FK
int	approved_by	FK

Diagram 3b — Deal lifecycle — conversion, reroute, earnings receivable and offset, balance conversion, entitlement re-attribution

3c · Settlement, return, confirmation, fees, shares, rebates, loss events and recoveries

Diagram 3c. `PROJECT`, `RECEIVER`, `MARKUP_SHARE_RULE` are defined in Diagram 1a; `PARTNER_CONFIG`, `SETTLEMENT_SENDING_ENTITY`, `SETTLEMENT_RAIL`, `DESTINATION_APPROVAL`, `PARTNER_REBATE_RULE` in 1b; `MARKET_RATE` in 2; `DEAL`, `COLLECTION`, `INVOICE` in 3a; `BALANCE_CONVERSION` in 3b.


SETTLEMENT_RAIL
SETTLEMENT_SENDING_ENTITY
PARTNER_CONFIG
PROJECT
BALANCE_CONVERSION
MARKUP_SHARE_RULE
DEAL
PARTNER_REBATE_RULE
MARKET_RATE
RECEIVER
INVOICE
DESTINATION_APPROVAL
COLLECTION
rail
narrative
draws from this partner's holding
client
funds a return leg (funding_source = returned_settlement)
rejected and returned
applied to
post-confirmation true-up / bounce fee
markup
network / bank fee override (subject = Settlement)
markup on a return leg (subject = Deal)
on this deal
accrues per deal
on this deal
accrues per deal
leg settled by this line (required for group-less receivers)
one or many destinations
discharge snapshot
discharged by
paid receiver (offset if in another group)
destination account
topped up by
short receiver
confirmed
loss on a released line
loss on a verified part
when short
recovered share by share

SETTLEMENT

int	id	PK
int	project_id	FK
int	partner_config_id	FK
int	settlement_sending_entity_id	FK
int	settlement_rail_id	FK
string	currency	
decimal	total	
string	bank_fee_treatment	
string	state	
int	prepared_by	FK
int	approved_by	FK
int	escalated_to	FK
datetime	released_at	
datetime	cutoff_at	
date	execution_due_date	

SETTLEMENT_RETURN

int	id	PK
int	settlement_line_id	FK
decimal	returned_amount	
decimal	bank_fee	
string	reason	
datetime	returned_at	

BANK_FEE_EVENT

int	id	PK
int	settlement_id	FK
decimal	expected_amount	
decimal	actual_amount	
string	treatment	
string	state	
int	applied_settlement_id	FK

FEE_DECISION

int	id	PK
string	subject_type	
int	subject_id	
string	fee_kind	
string	decision	
decimal	pct	
int	recommended_by	FK
int	approved_by	FK
string	reason	

MARKUP_SHARE_ACCRUAL

int	id	PK
int	markup_share_rule_id	FK
int	deal_id	FK
decimal	amount	
string	state	
datetime	paid_at	
string	direction	
string	period	
decimal	received_amount	

PARTNER_REBATE_ACCRUAL

int	id	PK
int	partner_rebate_rule_id	FK
int	deal_id	FK
decimal	expected_amount	
decimal	received_amount	
string	period	
string	state	
string	statement_ref	

SETTLEMENT_LINE

int	id	PK
int	settlement_id	FK
int	deal_id	FK
int	destination_approval_id	FK
int	paid_receiver_id	FK
decimal	amount	
decimal	receiver_fee_applied	
string	reference	
string	outcome	
int	invoice_id	FK
decimal	obligation_discharged	
decimal	discharge_rate	
int	discharge_market_rate_id	FK
string	token_ref	
string	evidence_ref	

CONFIRMATION

int	id	PK
int	settlement_line_id	FK
decimal	amount_received	
string	slip_ref	
string	result	
int	confirmed_by	FK
datetime	confirmed_at	

LOSS_EVENT

int	id	PK
string	side	
int	collection_id	FK
int	settlement_line_id	FK
decimal	amount	
string	reason	
datetime	occurred_at	
decimal	sender_share	
decimal	npl_share	
decimal	partner_share	
decimal	client_share	
string	state	
int	approved_by	FK
decimal	pool_funded	
decimal	wallet_funded	
string	funding_reference	

SHORTFALL

int	id	PK
int	receiver_id	FK
int	confirmation_id	FK
decimal	amount	
string	state	
int	topped_up_settlement_id	FK

LOSS_RECOVERY

int	id	PK
int	loss_event_id	FK
string	party	
decimal	amount	
string	state	
datetime	received_at	
string	reference	
int	approved_by	FK

Diagram 3c — Settlement, return, confirmation, fees, shares, rebates, loss events and recoveries

Derived views (never stored)

View	Definition
`PROJECT_BALANCE(project, partner, currency)`	PROJECT_BALANCE(project, partner, currency) = the balance of CL.<PARTNER>.<CCY>.DUE for the partner configuration: Σ debits – Σ credits over postings — conversions in, top-ups, make-goods, offsets and hops in, releases and offsets out — never a sum over deals (the definition summed conversions' actual_out, which includes NPL's margin, and had no term for the events). Was CLIENT_BALANCE. Also reports a pending column — the expected_out of legs whose rate is locked with the partner but whose money has not yet been collected or converted; a void removes it; pending is never a posting.
`GROUP_ENTITLEMENT(group, currency)`	GROUP_ENTITLEMENT(group, currency) = Σ credits – Σ debits over CL.PROJECT.<CCY>.PAYABLE postings carrying receiver_group_id = the group: promises raised at collection and conversion, re-raised on a return or a short confirmation, re-denominated by an offset or a balance conversion, reduced by a release or a client loss share. A payout to a group-less receiver carries the paying leg's group (SETTLEMENT_LINE.deal_id). Σ over a project's groups = the PAYABLE balance = Σ client assets, per currency.
`OFFSET(group, currency)`	OFFSET(group, currency) = GROUP_ENTITLEMENT; positive = owed, negative = holding another group's flow. Arises only when a counterparty receiver is paid from a leg attributed to another group; zero by construction in a single-group project. Carries Management approval and tighter aging.
`CUSTODY(own_wallet)`	CUSTODY(own_wallet) = Σ collections into the own wallet under open reroutes + Σ recovered dues resting there + Σ own-desk client balances + Σ bridge collections not yet forwarded – Σ forwarded or paid out; each row with deal, reason, role and age. The standing control on money in NPL's hands.
`INVOICE_BALANCE(invoice)`	INVOICE_BALANCE(invoice): for a vendor invoice, INVOICE.amount – Σ SETTLEMENT_LINE.obligation_discharged over released lines not returned (a payment discharges on release); for a resale invoice, INVOICE.amount – Σ DEAL_GROUP.invoice_amount over the groups that collected it with a verified collection. Σ VENDOR_PAYABLE = Σ vendor-invoice balances; Σ RESALE_RECEIVABLE = Σ resale-invoice balances (
`PROJECT_MONTH_VOLUME(project, month, currency)`	PROJECT_MONTH_VOLUME(project, month, currency) = Σ DEAL.amount_in over forward legs of the project collected in the calendar month, in the tier's volume currency. Read at quote to choose the FEE_TIER; stamped on the leg as tier_volume_basis.
`CLIENT_POSITION(client, currency)`	CLIENT_POSITION(client, currency) = Σ over the client's projects of PROJECT_BALANCE (per partner), GROUP_ENTITLEMENT, CUSTODY, and the period's earnings (EARN_GROSS) and rebates (EARN_REBATE) from postings — the client-level report that project = client made impossible. Compared with CLIENT.exposure_ceiling when set. Its one-currency total is a REPORTING_VALUE, an estimate at the latest market rate.

View	Definition
`WALLET_REGISTRY(network, address)`	WALLET_REGISTRY(network, address) = every wallet address the model knows, with its owner and role: the partners' receiving wallets (COLLECTION_RECEIVING_ENDPOINT kind wallet, with risk_band), NPL's own wallets (OWN_WALLET with role), receivers' wallets (RECEIVING_ENTITY_ACCOUNT kind wallet, with their DESTINATION_APPROVALs) and the sender addresses seen in WALLET_SCREENING with their latest decision. A view, never a second copy of an address: it answers which networks a partner accepts, which wallet is the bridge, and whether an address was seen before under another owner — the FRS 'wallet registry'.
`REPORTING_VALUE(subject, currency, as_of)`	REPORTING_VALUE(subject, currency, as_of) = any per-currency figure of the model — a PROJECT_BALANCE, a CLIENT_POSITION line, a month's collection volume (PROJECT_MONTH_VOLUME), earnings and rebates from postings, custody — multiplied by the latest MARKET_RATE of the platform pair for that currency (its PAIR_RATE_SOURCE chain), with the market_rate_id and captured_at it used and a stale flag when the rate is older than reporting_rate_max_age_min. An estimate for display: the figures in each currency stay the facts, the converted total is a view over them at the moment it is read. Serves project-level and overall collection volumes, earnings and current balances in one currency.
`RESELLER_MARGIN(resale_invoice)`	RESELLER_MARGIN(resale invoice) = resale INVOICE.amount − the amount of INVOICE.vendor_invoice_id, in the invoice currency, equal to RESELL_REVENUE − RESELL_COST posted for the pair: NPL-GR's game reseller revenue, fixed when the two invoices are posted and untouched by anything the remittance side does (;
`REMITTANCE_MARGIN(deal)`	REMITTANCE_MARGIN(deal) = EARN_GROSS (net of markup shares) − EXP_PARTNER ± VAR_FX_TIMING − EXP_BANKFEE − EXP_NETWORK − EXP_ROUNDING on the deal's postings, in the converted currency: NPL's remittance earnings on one deal, including the forex result of instalments paid at different days' rates.
`NPL_PROFIT(scope, period, currency)`	NPL_PROFIT(scope, period, currency) = remittance earnings (EARN_GROSS + EARN_REBATE − EXP_PARTNER − EXP_BANKFEE − EXP_NETWORK − EXP_ROUNDING − EXP_LOSS ± VAR_CONVERSION ± VAR_CUTOFF ± VAR_RATE_HONOUR ± VAR_FX_TIMING) + reseller earnings (RESELL_REVENUE − RESELL_COST), each callable on its own and together, by project, partner, pair or month, per currency; one figure across currencies is a REPORTING_VALUE.

Notes

- **Every deal belongs to a deal group.** `DEAL_GROUP` is the client-facing instruction: the pair as the sender sees it, the sender rate, fee, reference and entry mode. A plain single-partner deal is a group of one leg; a route that needs two partners (INR → USDT at Ali, USDT → EUR at Jeton) has two forward legs; a rejected payout is unwound with a return leg.
- **Legs.** `leg_type` forward or return; `funding_source` collection, prior_leg (leg 2 of a route, no collection) or returned_settlement (a return leg). Priced once and partner fixed at quote hold per leg. Attribution is to the receiver group: set at quote on the first forward leg, inherited by leg n of a route from leg n−1 and by a return leg from its parent leg. The receiver is optional until settlement.
- Two entry modes as before: collect-first legs are born in state Collected with priced_at = conversion time; quote-first legs carry sent / valid / grace timestamps and a late decision. An estimate-first mode

gives an indicative figure and prices at conversion.

- **Conversion has exactly one owner**: a deal leg or a `BALANCE_CONVERSION`. The receiver-side fee is taken at conversion. Amounts are rounded at conversion by the pair's option; a dropped fraction (`rounding_residual`, signed) stays client money in the balance and pays out when it reaches a unit, an amount added by rounding up is NPL's cost. NPL's margin is captured at conversion and is never again at risk from a return or a forward.
- **Balance conversion** is NPL converting a client's held balance (5K SGD → USDT to fund a 15K USDT payout): Finance-initiated, Management-approved, no sender, receiver or collection, same rate machinery, markup by `FEE_DECISION`. It names the receiver group it serves; `ENTITLEMENT_REATTRIBUTION` moves that group's entitlement to the new currency, and any excess over the group's entitlement is booked as an approved offset.
- **Reroute** is the only way partner or partner entity change after quote. Forwarding is forward_full or forward_and_recover_dues: NPL nets what the alternate partner owes it (`EARNINGS_RECEIVABLE`) against the funds forwarded, within the outstanding receivable, with the partner's agreement recorded. Post-collection outcome is via the own wallet or a refund to the sender.
- One settlement model: `SETTLEMENT` drains one partner's holding into one or many `SETTLEMENT_LINE`s, rounded as the pair says. Under settlement_policy to_zero the balance after release is less than one unit; under hold_allowed the balance may rest at the partner within the project's exposure cap.
- A line paying a counterparty receiver whose group differs from the paying leg's attributed group is an offset, created by the same action, with Management approval and tighter aging. A line paying a sender_return or partner_transit receiver is never an offset: those receivers have no group, and the line debits the group of the leg it names in `deal_id`.
- **Rejected payout.** When the receiving or intermediary bank rejects a transfer, `SETTLEMENT_RETURN` records the returned amount, bank fee and reason; the line's outcome becomes returned, a reversing ledger transaction puts the money back into the client's balance and the paying leg's group entitlement, and `BANK_FEE_EVENT` books the bounce fee. The money then stays in balance for later payouts, or funds a return leg — attributed to the same group — that converts back and pays the sender's return receiver.
- **Fee decisions.** `FEE_DECISION` is a one-off, approved decision on a single subject: markup applied or waived on a return leg or balance conversion, a network-fee override on a refund, a bank-fee treatment override on one settlement. Standing rules stay in `FEE_STRUCTURE` and `FEE_OVERRIDE`.
- `CONFIRMATION` is full or short; a short confirmation opens a `SHORTFALL` topped up by a later settlement or an explicit separate payout — cash found short at hand-over included. Rounding differences no longer land here. `MARKUP_SHARE_ACCRUAL` and `PARTNER_REBATE_ACCRUAL` accrue per deal under their rules; a party_retains markup-share accrual is a receivable from the party, reconciled monthly like a rebate; rebates are reconciled monthly against the partner's statement, which covers NPLify deals only.
- **Pricing stamps.** A leg stamps the override and the base tier that priced it; a leg that crosses a tier threshold holds two `DEAL_FEE_TIER` portions and a blended fee. The deal group stamps the customer-facing fee and rate the quote printed. On a pair that needs a bank lock the leg records `rate_locked_at` and `conversion_due_date`: locked before the partner's cutoff it converts the same day, otherwise the next business day at that day's version; the sender's figures are write-once, NPL absorbs the difference as a cutoff-timing variance, and a conversion whose next-day rate breaches the margin is deferred with a reason. The client balance view shows the locked leg's converted value as pending from the lock; a void removes it. `QUOTE_PACKAGE` is the record of what was sent for a deal group — template version, channel, printed figures, `sent_at`, hash — one row per issue; a quote-first leg is Sent only when a package has sent_at, re-issues are new rows, a new price is a new deal group.

- **Invoices.** `INVOICE` is an obligation in its own currency — a vendor's EUR invoice a reseller customer pays in THB or USDT, in parts, over days. A deal group names the invoice it pays (advance, then balance); each payout line records how much of the obligation it discharged and the market snapshot used; `INVOICE_BALANCE` is the open remainder.
- **Cash and crypto parts.** A cash part carries the token (the local-currency bill whose serial identifies the agent), the agent's contact and the partner's count; the leg carries the city, which also selects the rate version. A crypto part carries the sender's declared address and the screening that admitted it, unless the project skips up-front screening.
- **Loss events.** `LOSS_EVENT` records money lost after it was verified or released — a frozen local account, cash short that cannot be recovered, a wrong-chain transfer — on either side, by any method, for a partial or full amount, with the loss split across sender, NPL, partner and client. The whole loss is booked to NPL and the client made whole in one transaction (source LossEvent; MAKEGOOD / LOSS_PAYABLE when it cannot be funded at once); Management approves the split and each share is a separate `LOSS_RECOVERY` posting; the collection part stays verified and is reversed only when the deal is unwound.

4 · Ledger & controls domain

Diagram 4. `OWN_WALLET`, `RECEIVER`, `SENDER`, `MARKUP_SHARE_PARTY` are defined in Diagram 1a; `PARTNER_CONFIG` in 1b; `COLLECTION`, `INVOICE` in 3a; `CONVERSION`, `BALANCE_CONVERSION`, `REROUTE`, `EARNINGS_OFFSET` in 3b; `SETTLEMENT`, `SETTLEMENT_RETURN`, `CONFIRMATION`, `BANK_FEE_EVENT`, `MARKUP_SHARE_ACCRUAL`, `PARTNER_REBATE_ACCRUAL`, `LOSS_EVENT`, `LOSS_RECOVERY` in 3c.

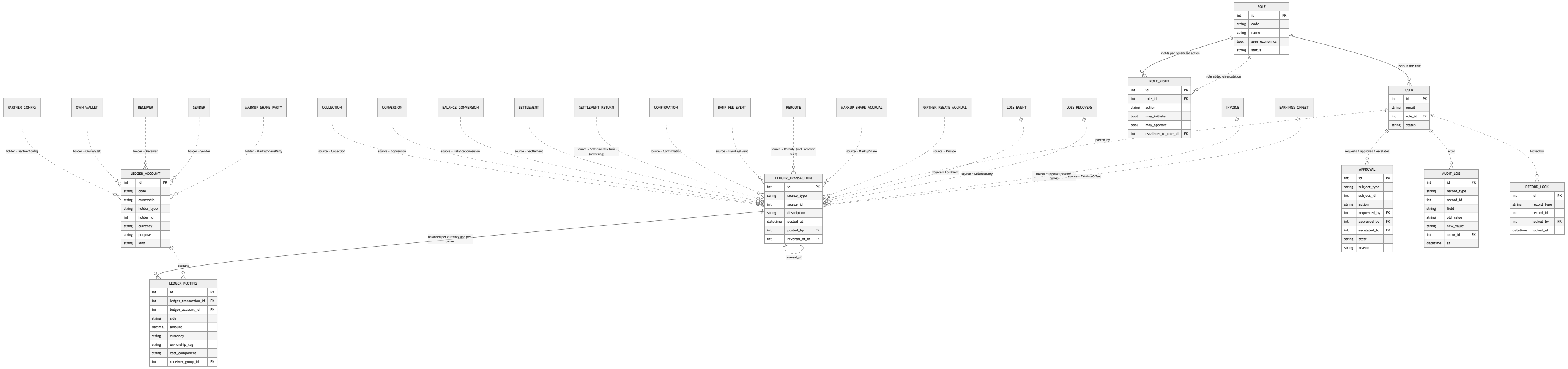

PARTNER_CONFIG
OWN_WALLET
RECEIVER
SENDER
MARKUP_SHARE_PARTY
COLLECTION
CONVERSION
BALANCE_CONVERSION
SETTLEMENT
SETTLEMENT_RETURN
CONFIRMATION
BANK_FEE_EVENT
REROUTE
MARKUP_SHARE_ACCRUAL
PARTNER_REBATE_ACCRUAL
LOSS_EVENT
LOSS_RECOVERY
INVOICE
EARNINGS_OFFSET
holder = PartnerConfig
holder = OwnWallet
holder = Receiver
holder = Sender
holder = MarkupShareParty
source = Collection
source = Conversion
source = BalanceConversion
source = Settlement
source = SettlementReturn (reversing)
source = Confirmation
source = BankFeeEvent
source = Reroute (incl. recover dues)
source = MarkupShare
source = Rebate
source = LossEvent
source = LossRecovery
source = Invoice (reseller books)
source = EarningsOffset
posted_by
LEDGER_ACCOUNT
int id PK
string code
string ownership
string holder_type
int holder_id
string currency
string purpose
string kind
account
balanced per currency and per owner
LEDGER_POSTING
int id PK
int ledger_transaction_id FK
int ledger_account_id FK
string side
decimal amount
string currency
string ownership_tag
string cost_component
int receiver_group_id FK
LEDGER_TRANSACTION
int id PK
string source_type
int source_id
string description
datetime posted_at
int posted_by FK
int reversal_of_id FK
reversal_of
ROLE
int id PK
string code
string name
bool sees_economics
string status
rights per controlled action
role added on escalation
users in this role
ROLE_RIGHT
int id PK
int role_id FK
string action
bool may_initiate
bool may_approve
int escalates_to_role_id FK
USER
int id PK
string email
int role_id FK
string status
requests / approves / escalates
actor
locked by
APPROVAL
int id PK
string subject_type
int subject_id
string action
int requested_by FK
int approved_by FK
int escalated_to FK
string state
string reason
AUDIT_LOG
int id PK
string record_type
int record_id
string field
string old_value
string new_value
int actor_id FK
datetime at
RECORD_LOCK
int id PK
string record_type
int record_id
int locked_by FK
datetime locked_at

Diagram 4 — Ledger & controls

Notes

- Client due balances are held as one account per project × partner × currency (CL.<PARTNER>.<CCY>.DUE); group entitlement is a view, not an account. `holder_type` never includes a settlement rail or narrative entity.
- **Ownership moves between client and company only at named events** — margin recognition at conversion, the two sides of an earnings offset, the forex true-ups of invoice lines, fees NPL absorbs or covers, top-ups, make-wholes, make-goods and the client's loss share — and every transaction balances per owner as well as per currency. NPL never advances money to a receiver, and withholding on a reroute is not a separate event because the margin is always captured by whichever partner converts.
- Recover dues on a reroute posts own-wallet cash against the earnings receivable at the partner: company money on both sides, no income.
- Transaction sources: Collection, Conversion, BalanceConversion, Settlement, SettlementReturn, Confirmation, BankFeeEvent, Reroute, MarkupShare, Rebate, LossEvent, LossRecovery, Invoice, EarningsOffset, Adjustment. Account purposes include PAYABLE, SHARE_PAYABLE, SHARE_RECEIVABLE, REBATE_RECEIVABLE, LOSS_RECEIVABLE, MAKEGOOD / LOSS_PAYABLE, EARN_GROSS, EARN_REBATE, the four VAR_ accounts, FX_RESULT, the reseller purposes (RESELL_REVENUE, RESELL_COST, VENDOR_PAYABLE, RESALE_RECEIVABLE, REMIT_CLAIM), EXP_ROUNDING and EXP_LOSS; ADVANCE and INTRANSIT do not exist. Cost components include share, rebate, rounding, fx_timing, cutoff_timing, loss and resale.
- **Three streams NPL never mixes.** Earnings on deals post to EARN_GROSS at conversion; a partner's rebate posts to EARN_REBATE and is never in the base a client party shares; NPL's remainder from a party that holds the converted funds sits in SHARE_RECEIVABLE until the party pays it. The cost of rounding up is EXP_ROUNDING. A loss is booked to EXP_LOSS in full first and the client made whole; the partner's and the sender's agreed shares are LOSS_RECEIVABLE owed to NPL, the client's share is a partial reversal of the make-good, and what remains in EXP_LOSS is NPL's own share. Own-desk deals use the ordinary partner-coded accounts with the own-desk configuration as holder and EXP_PARTNER zero by construction.
- One system-settings area: `SYSTEM_SETTING` holds NPL-wide, Management-approved keys — the reporting currency, its staleness limit, the operating time zone; `ROLE` holds the user roles and `ROLE_RIGHT` what each may initiate and approve per controlled action, with the role added on escalation; platform-scope `CURRENCY_PAIR`s carry the conversion rates through the ordinary rate-source chain. `REPORTING_VALUE` converts any per-currency figure into the reporting currency at the latest market rate of its platform pair, as a display-only estimate that carries the rate it used. `APPROVAL`, `AUDIT_LOG` and `RECORD_LOCK` are modelled once and apply to all master data. Approver ≠ initiator on every approval; Operations never sees derived economics. Approval actions include price_quote, extend_quote_validity, refund_to_third_party, cross_group_payout, balance_conversion, recover_dues, cover_bank_fee, raise_fee_cap, enable_own_desk, approve_loss_split, override_screening_reject and earnings_offset (ROLE_RIGHT rows). cross_group_payout never fires on a line paying a sender_return or partner_transit receiver. Wallet screening itself is an Operations action, not an approval.

Appendix A · FRS reconciliation

A.1 Vocabulary

FRS term	Model entity
Client	`CLIENT (the commercial relationship) — its arrangements are PROJECTs (until the FRS client mapped to PROJECT)`
Project (NPL's 'Project Context': BF, BF Sub, Evo …)	`PROJECT under CLIENT — one arrangement with its own fees, partners and policies`
Receiving Entity	`RECEIVING_ENTITY (member of a counterparty RECEIVER, which belongs to a RECEIVER_GROUP) with its RECEIVING_ENTITY_ACCOUNTs`
Sender	`SENDER with SENDER_RECEIVER_ALLOW; its refund destination is a group-less sender_return RECEIVER`
Partner (PT, Jeton)	`PARTNER + PARTNER_CONFIG`
Facilitating Entity / Wallet registry	`COLLECTION_RECEIVING_ENDPOINT (kind wallet) under PARTNER_ENTITY for the partner's wallets; the FRS 'wallet registry' across all owners is the WALLET_REGISTRY view`
Rate-day versions	`RATE_COMPARISON + PARTNER_RATE_VERSION`
Settlement	`SETTLEMENT + SETTLEMENT_LINE + CONFIRMATION (+ SETTLEMENT_RETURN when rejected)`
Shortfall ledger	`SHORTFALL + CL.<RECEIVER>.<CCY>.SHORTFALL`
PT Comms / H Comms	`CO.<PARTNER>.<CCY>.POOL (earnings receivable)`
Inter-entity redirection	`Offset across receiver groups: a counterparty paid_receiver_id in another group than the paying leg's`
Double conversion / pass-through	`DEAL_GROUP with two legs (funding_source = prior_leg), hop via a group-less partner_transit RECEIVER`
Commission / rebate (Jeton 0.4 %, Aquanow 0.3 %)	`PARTNER_REBATE_RULE (basis pct_of_amount in payout_currency, scoped by sender or fee structure) + PARTNER_REBATE_ACCRUAL; income in EARN_REBATE`
Markup shared with a customer (Raeen sub-accounts, downlines)	`MARKUP_SHARE_RULE basis share_of_net_markup, party_kind client_party, settlement_mode party_retains; SHARE_RECEIVABLE`
Fee by monthly volume (GDC)	`FEE_TIER under FEE_STRUCTURE; DEAL_FEE_TIER portions; PROJECT_MONTH_VOLUME`
Game reseller / NPL-GR	`PROJECT with an own-desk PARTNER_CONFIG; OWN_WALLET as COLLECTION_RECEIVING_ENDPOINT; INVOICE paid in parts`
Customer-facing calculation ('market rate', 'fee 1 %')	`FEE_STRUCTURE.customer_facing_fee_basis / pct; DEAL_GROUP.customer_facing_rate`
Token (cash collection / settlement)	`COLLECTION.token_ref, agent_contact_ref, counted_at; SETTLEMENT_LINE.token_ref; DEAL.cash_location; PARTNER_RATE_VERSION.location`
Bank cutoff (2.30 pm GMT+8)	`PARTNER.bank_cutoff_time + PARTNER_CUTOFF_OVERRIDE (extensions, early closes); PARTNER_PAIR.needs_bank_lock; DEAL.rate_locked_at, cutoff_at, conversion_due_date; SETTLEMENT.cutoff_at, execution_due_date`

FRS term	Model entity
Wallet screening	`WALLET_SCREENING; COLLECTION.sending_address; COLLECTION_RECEIVING_ENDPOINT.risk_band`
Reversal of collection; cash short	`LOSS_EVENT with its split; SHORTFALL for recoverable cash shortages`
Holiday calendar	`PARTNER.holiday_calendar (the calendar's name) + HOLIDAY rows (the dates); business-day arithmetic for due dates`
Quote package	`QUOTE_PACKAGE under DEAL_GROUP — one row per issue, hash = DEAL.quote_hash, sent_at starts validity`
Message template	`MESSAGE_TEMPLATE — versioned, global or per project, sender-facing fields only for quotes`
Settlement (FRS) — was DISBURSEMENT	`SETTLEMENT, SETTLEMENT_LINE, SETTLEMENT_RETURN; PROJECT.settlement_policy`
Markup shared with a party — was INTRODUCER / REFERRAL_*	`MARKUP_SHARE_PARTY, MARKUP_SHARE_RULE, MARKUP_SHARE_ACCRUAL; purpose SHARE_PAYABLE`
Approved destination — was SETTLEMENT_REGISTRATION	`DESTINATION_APPROVAL`
Sender pays / receiver pays; lowest and highest markup; customer-facing calculation; fee on top of market / fee inside its rate; agent rate	`sender_pays_pct, receiver_pays_pct; min_markup_pct, max_markup_pct; customer_facing_fee_basis / _pct / _rate; partner_fee_pct with fee_on_market / fee_in_rate; rate_basis market / agent_rate`
Client balance at a partner — was CLIENT_BALANCE	`PROJECT_BALANCE(project, partner, currency); the client-level roll-up is CLIENT_POSITION`
System / reporting currency	`SYSTEM_SETTING reporting_currency + platform-scope CURRENCY_PAIRs (their PAIR_RATE_SOURCE) + REPORTING_VALUE view, an estimate at the latest market rate`
User roles (Operations, Finance, Management)	`ROLE rows referenced by USER.role_id, with one ROLE_RIGHT row per role × controlled action (initiate / approve / escalate); rights per role in the Roles & Visibility Matrix`

A.2 Deviations from the FRS (classified)

FRS statement	Model	Classification
P1 — no custody	No custody except under an approved reroute or in an own-desk project (NPL-GR); OWN_WALLET + REROUTE for contingency, and the custody view as the control	Decided (answered;, 9 October 2026)
P3 — partner books never mix	Accounts per partner never net; alternate partner on reroute; post-collection transfer only if confirms	Decided (answered: reroute with recover dues; partner accounts never net)
Strict Client → Entity → Sender, one parent each	RECEIVER_GROUP → RECEIVER → RECEIVING_ENTITY → account; SENDER with a group allow-list; FRS project = one default group	Generalisation per NPL instruction
Beyond grace → re-quote always	late_collection_policy per project; FRS project keeps requote	Per NPL instruction (Sept 2026)

FRS statement	Model	Classification
Sender always priced on Kraken	FEE_STRUCTURE.rate_basis; FRS project = market	Generalisation
One settlement per deal, full amount	Accumulate → settle the balance rounded as the pair says (residual carried), or hold within the exposure cap where the project allows	Generalisation
Partner fixed at quote	Same, except under REROUTE	Consistent
Bank fee → shortfall → top-up	carry_forward is the FRS behaviour; waived / absorbed added	Per NPL instruction
Registry-only destinations	DESTINATION_APPROVAL.approval_status — FRS project has all approved	Generalisation
AQN-LT-Sub offset-settled collections	OWN_WALLET at Aquanow; pass-through is a two-leg deal group or a reroute via the own wallet; also NPL-GR's own-desk collection endpoint	Resolved

Appendix B · Data dictionary (field annotations)

Every table also carries `created_at` and `created_by`. Types: int, string, bool, decimal, datetime, date.

B.1 Configuration

PROJECT

Field	Type	Key	Notes
`id`	int	PK	
`client_id`	int	FK	the client this arrangement belongs to
`name`	string		
`client_label`	string		
`quote_freshness_min`	int		
`quote_validity_min`	int		
`grace_min`	int		
`grace_drift_pct`	decimal		
`late_collection_policy`	string		requote / honor_with_approval
`bank_fee_treatment_default`	string		waived / carry_forward / absorbed
`receiver_fee_timing`	string		at_conversion (default) / post_confirmation
`network_fee_policy`	string		absorbed / charged_back (project default; per-transaction override via FEE_DECISION). A network fee may or may not apply; when it does it is stated in USDT, when it does not it is zero and nothing posts
`settlement_policy`	string		to_zero / hold_allowed
`shortfall_tolerance`	decimal		
`exposure_ceiling`	decimal		cap on balance resting at a partner
`status`	string		

PARTNER

Field	Type	Key	Notes
`id`	int	PK	
`name`	string		Ali, Jeton, PT
`disclosure_style`	string		fee_in_rate / fee_on_market — whether the partner's fee sits inside its rate or is stated on top of market
`rate_expiry_time`	string		
`holiday_calendar`	string		name of the source calendar (e.g. ID banking holidays); the dates are HOLIDAY rows
`supported_methods`	string		crypto / bank / cash / local_deposit — the methods this partner offers (Ali: all four; Jeton and Aquanow: crypto, bank). Networks are implied by its wallet endpoints
`status`	string		
`bank_cutoff_time`	time		nullable: the partner's standing daily bank cutoff for pairs that rely on the banking channel (Ali and other Asian partners, 14:30); empty for Jeton and Aquanow, which bind nothing. Governs locking the rate for same-day conversion and initiating an outgoing bank settlement; per-day changes in PARTNER_CUTOFF_OVERRIDE
`cutoff_timezone`	string		IANA zone, e.g. Asia/Hong_Kong

RATE_SOURCE

Field	Type	Key	Notes
`id`	int	PK	
`code`	string		KRAKEN / CMC / OANDA / XE
`staleness_threshold_s`	int		

MARKUP_SHARE_PARTY

Field	Type	Key	Notes
`id`	int	PK	
`name`	string		
`status`	string		

SENDER

Field	Type	Key	Notes
`id`	int	PK	
`project_id`	int	FK	
`name`	string		Sender A, Sender B
`risk_tier`	string		
`status`	string		

RECEIVER_GROUP

Field	Type	Key	Notes
`id`	int	PK	
`project_id`	int	FK	
`name`	string		
`is_default`	bool		exactly one per project
`status`	string		

RECEIVER

Field	Type	Key	Notes
`id`	int	PK	
`project_id`	int	FK	
`group_id`	int	FK	receiver group; null unless kind = counterparty
`kind`	string		counterparty / sender_return / partner_transit
`sender_id`	int	FK	when kind = sender_return
`partner_config_id`	int	FK	when kind = partner_transit
`name`	string		the group: Receiver X, Receiver Y
`status`	string		

OWN_WALLET

Field	Type	Key	Notes
`id`	int	PK	
`project_id`	int	FK	
`provider`	string		e.g. Aquanow
`account_label`	string		e.g. LT Sub

Field	Type	Key	Notes
`role`	string		reroute_contingency / own_desk / collection_bridge — LT Sub at Aquanow is the NPL-GR own desk; NPL's ERC-20 wallet is a bridge for partners without that network
`networks`	string		networks the wallet receives on (ERC-20 where Ali cannot)
`network`	string		
`address`	string		
`signer_arrangement`	string		hardware / multisig
`custody_max_hours`	int		

THRESHOLD

Field	Type	Key	Notes
`id`	int	`PK`	
`project_id`	int	`FK`	
`kind`	string		max_amount / settlement_limit / daily_slots / rate_variation_rel / custom
`currency`	string		
`value`	decimal		
`condition_expr`	string		custom conditions
`escalation`	string		finance / management

CURRENCY_PAIR

Field	Type	Key	Notes
`id`	int	`PK`	
`project_id`	int	`FK`	null when scope = platform
`scope`	string		project / platform — a platform pair (currency → reporting currency) belongs to no project, is managed under system settings, is priced through PAIR_RATE_SOURCE like any pair and is never quoted
`from_currency`	string		
`to_currency`	string		
`quote_direction`	string		out_per_in / in_per_out
`rate_precision`	int		
`amount_rounding`	string		exact / truncate_unit / nearest_unit / round_up_unit / nearest_n / round_up_n (exact is the default for bank transfers — cents go out; the unit options apply where the pair says so, in practice to cash hand-overs)
`cash_rounding`	string		truncate_n / nearest_n / round_up_n — how the collected cash amount is rounded to cash_rounding_unit only)
`cash_rounding_unit`	int		100 / 500 / 1000 by currency — the unit the collected cash amount is rounded to; AED cash 100, USD cash in Dubai 10 (NPL, 9 Oct)
`status`	string		
`rounding_unit`	decimal		required for nearest_n / round_up_n (1,000 for THB, INR, JPY payouts)

PARTNER_CONFIG

Field	Type	Key	Notes
`id`	int	`PK`	
`project_id`	int	`FK`	
`partner_id`	int	`FK`	

Field	Type	Key	Notes
applicable_for_reroute	bool		
status	string		
is_own_desk	bool		default false; NPL converts in its own wallet for this project (NPL-GR); enabling needs Management approval

COLLECTION_SENDING_ENTITY

Field	Type	Key	Notes
id	int	PK	
sender_id	int	FK	
label	string		sender’s own account or a client’s account
account_ref	string		

SENDER_RECEIVER_ALLOW

Field	Type	Key	Notes
id	int	PK	
sender_id	int	FK	
receiver_group_id	int	FK	sendable_to

RECEIVING_ENTITY

Field	Type	Key	Notes
id	int	PK	
receiver_id	int	FK	
name	string		Entity X1, Entity X2
status	string		

PAIR_RATE_SOURCE

Field	Type	Key	Notes
id	int	PK	
currency_pair_id	int	FK	
rate_source_id	int	FK	
priority	int		

PARTNER_PAIR

Field	Type	Key	Notes
id	int	PK	
partner_config_id	int	FK	
currency_pair_id	int	FK	
partner_pricing	string		fee_on_market / fee_in_rate
partner_fee_pct	decimal		known only for fee_on_market
status	string		
needs_bank_lock	bool		the pair relies on the banking channel, so the partner’s cutoff governs its rate lock and its settlement (Evo pairs into EUR; USDT → USD / SGD at Ali); crypto-only pairs do not

SETTLEMENT_SENDING_ENTITY

Field	Type	Key	Notes
`id`	int	PK	
`partner_config_id`	int	FK	
`narrative_name`	string		what the receiver sees
`status`	string		

SETTLEMENT_RAIL

Field	Type	Key	Notes
`id`	int	PK	
`partner_config_id`	int	FK	
`name`	string		bank or EMI executing the transfer
`bank_details_ref`	string		
`expected_bank_fee`	decimal		
`currency`	string		

FEE_OVERRIDE

Field	Type	Key	Notes
`id`	int	PK	
`sender_id`	int	FK	
`partner_pair_id`	int	FK	
`pct`	decimal		
`effective_from`	datetime		
`approved_by`	int	FK	
`sender_pays_pct`	decimal		optional; with receiver_pays_pct replaces the structure’s proportion
`receiver_pays_pct`	decimal		optional; both blank → structure’s proportion; both set → sum = pct

RECEIVING_ENTITY_ACCOUNT

Field	Type	Key	Notes
`id`	int	PK	
`receiving_entity_id`	int	FK	
`currency`	string		
`rail`	string		SWIFT / SEPA / TRC-20 / ERC-20 / …
`account_kind`	string		bank / wallet
`bank_details_ref`	string		external store; when account_kind = bank
`network`	string		when account_kind = wallet
`address`	string		when account_kind = wallet; checksum-validated per network
`status`	string		
`usage`	string		standing / single_use — a local account the receiver gives for one payout
`reserved_for_deal_id`	int	FK	nullable; the one deal a single_use account serves

DESTINATION_APPROVAL

Field	Type	Key	Notes
`id`	int	PK	

Field	Type	Key	Notes
`partner_config_id`	int	FK	
`receiving_entity_account_id`	int	FK	
`approval_status`	string		pending_partner / approved / suspended
`bank_details_ref`	string		as paid by this partner; only when the account's kind is bank
`network`	string		as registered with this partner; only when the account's kind is wallet
`address`	string		as registered with this partner; only when the account's kind is wallet
`approved_at`	datetime		
`wallet_screening_id`	int	FK	nullable; the receiver-side screening of a wallet account

FEE_STRUCTURE

Field	Type	Key	Notes
`id`	int	PK	
`partner_pair_id`	int	FK	
`rate_basis`	string		market / agent_rate / partner)
`mode`	string		fixed / variable
`pct`	decimal		
`sender_pays_pct`	decimal		
`receiver_pays_pct`	decimal		
`min_markup_pct`	decimal		
`max_markup_pct`	decimal		
`min_margin_pct`	decimal		
`variance_threshold_rel`	decimal		
`quoting_rate_policy`	string		worst_applicable / designated / best_applicable
`effective_from`	datetime		
`customer_facing_fee_basis`	string		actual / sender_pays_on_agent_rate / fixed
`customer_facing_fee_pct`	decimal		when customer_facing_fee_basis = fixed (BF via Ali shows 1 %)

MARKUP_SHARE_RULE

Field	Type	Key	Notes
`id`	int	PK	
`markup_share_party_id`	int	FK	
`project_id`	int	FK	
`partner_pair_id`	int	FK	optional
`basis`	string		share_of_earnings / fixed_pct / share_of_net_markup (base = fee parts – partner cost on the leg, on amount × market; rebates never in the base)
`pct`	decimal		
`effective_from`	datetime		
`fee_structure_id`	int	FK	optional scope: only deals priced on this structure (Raeen sub-account 50 %, Raeen direct downlines 75 %)
`party_kind`	string		markup-share party / client_party
`settlement_mode`	string		npl_pays / party_retains — the party holds the converted funds and pays NPL the remainder monthly

PARTNER_REBATE_RULE

Field	Type	Key	Notes
`id`	int	PK	
`partner_config_id`	int	FK	
`effective_from`	datetime		
`basis`	string		pct_of_amount — the rebate is a percentage of the amount converted (Jeton 0.4 % of amount on a 1 % or 1.25 % fee, Novi 0.3 %, Aquanow 0.3 %)
`pct`	decimal		the rebate in the chosen basis
`fee_structure_id`	int	FK	optional scope
`sender_id`	int	FK	optional scope (Novi)
`payout_currency`	string		the currency the partner pays the rebate in and the accrual is expressed in — the amount currency

PARTNER_ENTITY

Field	Type	Key	Notes
`id`	int	PK	
`partner_config_id`	int	FK	
`name`	string		the partner’s legal vehicle: PT-tour, PT Global Inc
`status`	string		

COLLECTION_RECEIVING_ENDPOINT

Field	Type	Key	Notes
`id`	int	PK	
`partner_entity_id`	int	FK	the vehicle whose name is on the account or wallet
`kind`	string		bank / wallet
`currency`	string		
`rail`	string		nullable: a bank endpoint needs it to go active; optional for wallets, must match network when set
`bank_details_ref`	string		external store; when kind = bank
`network`	string		TRC-20 / ERC-20 / …; when kind = wallet
`address`	string		when kind = wallet; checksum-validated per network
`status`	string		
`own_wallet_id`	int	FK	nullable; set when the endpoint is NPL’s own wallet — under an own-desk configuration or as a collection bridge for a partner that lacks the network or for risk isolation; the endpoint still belongs to the leg’s partner configuration
`risk_band`	string		standard / elevated — which screening decisions may land here
`usage`	string		standing / single_use (ad hoc local deposit accounts)
`reserved_for_deal_id`	int	FK	nullable; the one deal a single_use endpoint serves
`expected_amount`	decimal		nullable; the amount a single_use endpoint for

FEE_TIER

Field	Type	Key	Notes
`id`	int	PK	
`fee_structure_id`	int	FK	
`min_monthly_volume`	decimal		threshold in volume_currency, per calendar month

Field	Type	Key	Notes
`volume_currency`	string		
`pct`	decimal		
`sender_pays_pct`	decimal		optional
`receiver_pays_pct`	decimal		optional
`effective_from`	datetime		

WALLET_SCREENING

Field	Type	Key	Notes
`id`	int	`PK`	
`project_id`	int	`FK`	
`subject_kind`	string		sender_wallet / receiver_wallet
`sender_id`	int	`FK`	when subject_kind = sender_wallet
`receiving_entity_account_id`	int	`FK`	when subject_kind = receiver_wallet
`network`	string		
`address`	string		
`provider_ref`	string		screening tool reference
`result`	string		standard / elevated / unacceptable
`decision`	string		accept / accept_alternate / reject
`screened_by`	int	`FK`	Operations
`screened_at`	datetime		
`evidence_ref`	string		

PARTNER_CUTOFF_OVERRIDE

Field	Type	Key	Notes
`id`	int	`PK`	
`partner_id`	int	`FK`	
`applies_on`	date		one day; null for a weekday rule
`weekday`	string		recurring rule (Friday); null for a one-day override
`cutoff_time`	time		the cutoff that applies instead
`kind`	string		extension / early_close
`reason`	string		urgent request / pre-holiday / partner notice
`partner_agreement_ref`	string		the partner agreed (extensions)
`recorded_by`	int	`FK`	Finance

HOLIDAY

Field	Type	Key	Notes
`id`	int	`PK`	
`partner_id`	int	`FK`	
`date`	date		
`name`	string		Idul Fitri, Boxing Day
`scope`	string		bank_channel / partner
`source`	string		partner notice / public calendar

MESSAGE_TEMPLATE

Field	Type	Key	Notes
`id`	int	PK	
`project_id`	int	FK	null = global default; a project row overrides it
`code`	string		quote / payment_instructions / confirmation_request / settlement_advice / cash_token
`version`	int		
`channel`	string		email / chat / portal / pdf
`language`	string		
`body_ref`	string		external content store
`printed_fields`	string		the model fields the template may print; sender-facing only for quote
`effective_from`	datetime		
`approved_by`	int	FK	Management
`status`	string		

CLIENT

Field	Type	Key	Notes
`id`	int	PK	
`name`	string		Betfair, Evo, 66 Group, GDC, NPL-GR
`category`	string		NPL's own grouping (Project Context tab: BF, Evo, Game Reseller …)
`legal_entity_ref`	string		contracting entity, external store
`kyc_ref`	string		external store
`relationship_owner`	int	FK	USER
`exposure_ceiling`	decimal		optional cap on Σ balance resting at partners across the client's projects, stated in the reporting currency and compared through REPORTING_VALUE at the latest market rate
`status`	string		

SYSTEM_SETTING

Field	Type	Key	Notes
`id`	int	PK	
`key`	string		reporting_currency / reporting_rate_max_age_min / operating_timezone / … (the reporting rate source is the platform pair's PAIR_RATE_SOURCE, not a key)
`value`	string		
`effective_from`	datetime		
`approved_by`	int	FK	Management

B.2 Rates

MARKET_RATE

Field	Type	Key	Notes
`id`	int	PK	
`rate_source_id`	int	FK	
`currency_pair_id`	int	FK	
`rate`	decimal		
`captured_at`	datetime		

RATE_COMPARISON

Field	Type	Key	Notes
`id`	int	PK	
`partner_pair_id`	int	FK	
`rate_date`	date		
`avg_spread_2w`	decimal		
`avg_spread_4w`	decimal		
`status`	string		open / closed
`created_at`	datetime		created lazily by the first deal of the day

PARTNER_RATE_VERSION

Field	Type	Key	Notes
`id`	int	PK	
`rate_comparison_id`	int	FK	
`version_no`	int		
`rate`	decimal		normalised to CURRENCY_PAIR.quote_direction; derived from quoted_rate
`market_rate_id`	int	FK	
`spread_pct`	decimal		
`status`	string		active / usable / breached / approved / rejected
`confirmed_by`	int	FK	
`approval_reason`	string		
`expires_at`	datetime		
`quoted_rate`	decimal		as the partner stated it (Ali: 1.1738 EUR→USDT style)
`quoted_direction`	string		out_per_in / in_per_out; rate = quoted_rate, or 1 ÷ quoted_rate at the pair's precision when the directions differ — derived, never typed
`location`	string		nullable; required on cash pairs — INR cash in Delhi and in Mumbai are two versions on one day

B.3 Deal, collection, conversion, reroute, balance conversion

DEAL_GROUP

Field	Type	Key	Notes
`id`	int	PK	
`project_id`	int	FK	
`sender_id`	int	FK	
`receiver_group_id`	int	FK	entitled group
`client_ref`	string		D-YYMMDD-seq, what the sender sees
`from_currency`	string		
`to_currency`	string		
`sender_rate`	decimal		
`fee_structure_id`	int	FK	
`amount_in`	decimal		
`target_out`	decimal		
`entry_mode`	string		quote_first / collect_first / estimate_first
`state`	string		

Field	Type	Key	Notes
quoted_at	datetime		
customer_facing_fee_pct	decimal		stamped at quote
customer_facing_rate	decimal		= sender_rate ÷ (1 − customer_facing_fee_pct); the ‘market rate’ the quote prints
invoice_id	int	FK	nullable; the resale invoice this group collects (NPL-GR) — the obligation the deal pays
invoice_amount	decimal		nullable; the part of the resale invoice this group collects, in the invoice currency — an advance, then the balance; Σ over the invoice’s groups ≤ its amount
invoice_rate	decimal		nullable; payout currency per invoice currency at pricing (same direction as SETTLEMENT_LINE.discharge_rate): the stored base for the amount reserved per instalment and for the forex true-up

DEAL

Field	Type	Key	Notes
id	int	PK	
deal_group_id	int	FK	every deal belongs to a group
leg_no	int		
leg_type	string		forward / return
funding_source	string		collection / returned_settlement / prior_leg
funding_ref_id	int		SETTLEMENT_RETURN id or prior DEAL id
project_id	int	FK	
sender_id	int	FK	
receiver_group_id	int	FK	attribution
receiver_id	int	FK	optional until settlement
partner_pair_id	int	FK	
partner_entity_id	int	FK	
amount_mode	string		target_out / fixed_in — fixed_in takes the fee off the principal, target_out adds it as a markup on the target
collection_method	string		crypto / bank / cash
cash_instructions	string		free text for the hand-over; the city is cash_location, the token is on the collection
fee_structure_id	int	FK	
market_rate_id	int	FK	the market snapshot the leg on; equals the earlier leg’s when rate_reused_from_deal_id is set
partner_rate_version_id	int	FK	
sender_rate	decimal		
amount_in	decimal		
gross_out	decimal		
client_net	decimal		
fee_sender_part	decimal		
fee_receiver_part	decimal		
priced_at	datetime		quote or conversion
sent_at	datetime		null on conversion-priced legs; = QUOTE_PACKAGE.sent_at of the package in force

Field	Type	Key	Notes
`valid_until`	datetime		end of validity as quoted; effective_valid_until is the one in force after extensions
`grace_until`	datetime		
`effective_valid_until`	datetime		the validity in force: valid_until as quoted, moved by each approved QUOTE_VALIDITY_EXTENSION (any number, no ceiling); grace_until follows it
`rate_reused_from_deal_id`	int	FK	nullable; this leg on the market rate and partner rate version of an earlier, expired leg at the sender's request — an approved override of the automated rate; the difference against the day's rate posts to VAR_RATE_HONOUR with component variance (the account named by, note corrected)
`rate_override_reason`	string		why the earlier rate
`rate_override_approved_by`	int	FK	Finance
`late_decision`	string		none / honored / requoted
`state`	string		
`quote_hash`	string		hash of the quote package in force for the group (QUOTE_PACKAGE.content_hash)
`fee_override_id`	int	FK	nullable; the override that priced this leg
`fee_tier_id`	int	FK	nullable; base tier at quote
`tier_volume_basis`	decimal		project month-to-date volume that justified the tier, stamped at quote
`rate_locked_at`	datetime		when the partner confirmed the rate; before the cutoff → converts the same day, after → next business day
`conversion_due_date`	date		the lock date when rate_locked_at precedes cutoff_at, else the next business day — weekends in the partner's cutoff_timezone and the partner's HOLIDAY rows skipped; Converting may not start earlier
`conversion_deferred_reason`	string		set when the due date is pushed because the next day's rate breaches the minimum margin (ask for a better rate, or wait a day)
`cash_location`	string		city of the cash hand-over; required on cash legs — rates and agents differ by city
`cutoff_at`	datetime		the effective cutoff that applied on the lock day (standing time or override), stamped with rate_locked_at

COLLECTION

Field	Type	Key	Notes
`id`	int	PK	
`deal_id`	int	FK	
`collection_receiving_endpoint_id`	int	FK	the partner endpoint paid into; null for cash
`collection_sending_entity_id`	int	FK	
`amount`	decimal		
`evidence_ref`	string		hash / slip / receipt
`state`	string		pending / verified / reversed; cash: token_issued → handed_over → counted → verified.: after a loss the part stays verified — NPL's funds stand in — and is reversed only when Management unwinds the deal
`verified_by`	int	FK	
`verified_at`	datetime		

Field	Type	Key	Notes
`sending_address`	string		crypto: the sender's declared wallet
`wallet_screening_id`	int	FK	nullable; the screening that admitted the sending wallet
`screening_skipped`	bool		projects that do not screen every sender up front (Evo)
`token_ref`	string		cash: serial of the token (local-currency bill) that identifies the agent; the partner may replace the agent and the token before hand-over — the collection holds the current token, the audit log the previous one; the rate locked at quote is unchanged
`agent_contact_ref`	string		cash: the receiving agent's contact, external store
`counted_at`	datetime		cash: the partner counted and confirmed the funds

CONVERSION

Field	Type	Key	Notes
`id`	int	PK	
`deal_id`	int	FK	null when owned by a balance conversion
`balance_conversion_id`	int	FK	null when owned by a deal leg
`partner_rate_version_id`	int	FK	
`calc_mode`	string		npl_calculates / partner_calculates
`amount_in`	decimal		
`expected_out`	decimal		
`actual_out`	decimal		booked figure, rounded as the pair says
`rounding_residual`	decimal		signed: positive = fraction dropped, stays client money; negative = amount added by rounding up, NPL's cost posted to EXP_ROUNDING
`fee_receiver_part`	decimal		taken at conversion
`variance`	decimal		
`residual_in`	decimal		the margin kept unconverted in the collection currency (NPL's practice: earnings are taken out of money not yet converted); EARN_GROSS is then recognised in that currency and rests in NPL's pool at the partner in that currency
`basis_variance`	decimal		
`state`	string		
`approved_by`	int	FK	

REROUTE

Field	Type	Key	Notes
`id`	int	PK	
`deal_id`	int	FK	
`trigger`	string		partner_unavailable / limit_reached / holiday (a HOLIDAY row of scope partner on the day) / compliance_hold / other
`phase`	string		pre_collection / post_collection
`outcome`	string		via_own_wallet / refund_sender (post-collection)
`own_wallet_id`	int	FK	
`alternate_partner_config_id`	int	FK	
`forwarding_policy`	string		forward_full / forward_and_recover_dues
`recovered_amount`	decimal		≤ outstanding receivable at the alternate partner

Field	Type	Key	Notes
partner_agreement_ref	string		the partner agreed the net amount
reason	string		
requested_by	int	FK	
approved_by	int	FK	

SENDER_CREDIT

Field	Type	Key	Notes
id	int	PK	
sender_id	int	FK	
deal_id	int	FK	
amount	decimal		
state	string		open / applied

EARNINGS_RECEIVABLE

Field	Type	Key	Notes
id	int	PK	
partner_config_id	int	FK	
deal_id	int	FK	
amount	decimal		
state	string		outstanding / recovered
recovered_by_deal_id	int	FK	
recovered_by_offset_id	int	FK	nullable; the EARNINGS_OFFSET that moved this margin to another partner’s unconverted balance
recovered_at	datetime		

BALANCE_CONVERSION

Field	Type	Key	Notes
id	int	PK	
project_id	int	FK	
partner_config_id	int	FK	
receiver_group_id	int	FK	group served; chosen split
from_currency	string		
to_currency	string		
partner_pair_id	int	FK	
partner_rate_version_id	int	FK	
market_rate_id	int	FK	
amount_in	decimal		
expected_out	decimal		
actual_out	decimal		
variance	decimal		
rounding_residual	decimal		
state	string		
requested_by	int	FK	Finance

Field	Type	Key	Notes
`approved_by`	int	FK	Management
`converted_at`	datetime		
`rate_locked_at`	datetime		
`conversion_due_date`	date		banking-ready balances (USD, SGD) wait for the next business morning

ENTITLEMENT_REATTRIBUTION

Field	Type	Key	Notes
`id`	int	PK	
`balance_conversion_id`	int	FK	
`receiver_group_id`	int	FK	
`amount_from`	decimal		
`amount_to`	decimal		

DEAL_FEE_TIER

Field	Type	Key	Notes
`id`	int	PK	
`deal_id`	int	FK	
`fee_tier_id`	int	FK	
`amount_in_portion`	decimal		Σ over a leg = amount_in
`pct`	decimal		the tier's pct at quote

INVOICE

Field	Type	Key	Notes
`id`	int	PK	
`project_id`	int	FK	
`kind`	string		vendor (the vendor's invoice to NPL-GR at the preferential price) / resale (NPL-GR's invoice to its customer at retail price)
`vendor_invoice_id`	int	FK	on a resale invoice: the vendor invoice it marks up (self-reference); RESELLER_MARGIN pairs the two through it
`receiver_id`	int	FK	the vendor owed, on a vendor invoice
`sender_id`	int	FK	the customer invoiced, on a resale invoice
`currency`	string		
`amount`	decimal		
`reference`	string		
`issued_at`	datetime		
`status`	string		open / settled / void

QUOTE_PACKAGE

Field	Type	Key	Notes
`id`	int	PK	
`deal_group_id`	int	FK	
`issue_no`	int		1, 2, … per deal group
`message_template_id`	int	FK	the template version that rendered it
`channel`	string		email / chat / portal / pdf

Field	Type	Key	Notes
`rendered_ref`	string		external document store
`content_hash`	string		= DEAL.quote_hash of the legs it prices
`printed_source_rate`	decimal		= DEAL_GROUP.customer_facing_rate
`printed_fee_pct`	decimal		= DEAL_GROUP.customer_facing_fee_pct
`printed_amount_out`	decimal		
`valid_until`	datetime		= the leg's valid_until
`sent_at`	datetime		marks the package sent; starts validity
`sent_by`	int	FK	Operations
`superseded_by_id`	int	FK	the re-issue that replaced this one, if any

QUOTE_VALIDITY_EXTENSION

Field	Type	Key	Notes
`id`	int	PK	
`deal_id`	int	FK	
`extension_no`	int		1, 2, … per leg
`requested_at`	datetime		
`from_valid_until`	datetime		
`to_valid_until`	datetime		
`reason`	string		the sender's request
`approved_by`	int	FK	Finance; Management past the cutoff or the rate version's expiry
`approval_id`	int	FK	

EARNINGS_OFFSET

Field	Type	Key	Notes
`id`	int	PK	
`project_id`	int	FK	
`from_partner_config_id`	int	FK	where NPL's converted margin rests
`from_currency`	string		
`from_amount`	decimal		NPL's margin handed to the client there
`to_partner_config_id`	int	FK	where the client's unconverted money is
`to_currency`	string		
`to_amount`	decimal		= from_amount ÷ the rate used, taken from the client's unconverted balance
`partner_rate_version_id`	int	FK	the to-partner's rate of the day (NPL, 11 Oct: the partner's rate, not market)
`market_rate_id`	int	FK	the market snapshot of the day, for reference
`requested_by`	int	FK	Finance
`approved_by`	int	FK	Management
`state`	string		requested / approved / posted
`posted_at`	datetime		

B.4 Settlement, return, confirmation, fees, shares, rebates, losses and recoveries

SETTLEMENT

Field	Type	Key	Notes
`id`	int	PK	
`project_id`	int	FK	
`partner_config_id`	int	FK	
`settlement_sending_entity_id`	int	FK	
`settlement_rail_id`	int	FK	
`currency`	string		
`total`	decimal		rounded as the pair says
`bank_fee_treatment`	string		
`state`	string		
`prepared_by`	int	FK	
`approved_by`	int	FK	
`escalated_to`	int	FK	
`released_at`	datetime		
`cutoff_at`	datetime		the effective cutoff of the approval day for a bank rail (standing time or override)
`execution_due_date`	date		the approval date when approved before cutoff_at, else the next business day (weekends and HOLIDAY rows skipped); a bank transfer is initiated no earlier

SETTLEMENT_LINE

Field	Type	Key	Notes
`id`	int	PK	
`settlement_id`	int	FK	
`deal_id`	int	FK	nullable; required when the paid receiver is group-less, optional on counterparty lines
`destination_approval_id`	int	FK	
`paid_receiver_id`	int	FK	
`amount`	decimal		
`receiver_fee_applied`	decimal		exceptional; fee is normally taken at conversion
`reference`	string		entity invoice format
`outcome`	string		sent / confirmed / short / returned — sent from release until the receiver's confirmation; short when the CONFIRMATION has result short and a SHORTFALL row re-raises the missing part; every released line ends confirmed, short or returned
`invoice_id`	int	FK	nullable; the invoice this line discharges — the vendor invoice for a vendor line, the resale invoice for the margin line to NPL-GR's own wallet
`obligation_discharged`	decimal		in the invoice currency; reserved = obligation_discharged × DEAL_GROUP.invoice_rate, paid = obligation_discharged × discharge_rate; reserved – paid is the forex true-up posted to VAR_FX_TIMING against NPL's pool
`discharge_rate`	decimal		payout currency per invoice currency at payment
`discharge_market_rate_id`	int	FK	nullable; the market snapshot used

Field	Type	Key	Notes
`token_ref`	string		cash settlement: the receiver's token, provided up front and relayed NPL → partner → agent
`evidence_ref`	string		cash settlement: the agent's photo of the receiver's token after hand-over, relayed partner → NPL → receiver; the receiver's CONFIRMATION closes the loop

SETTLEMENT_RETURN

Field	Type	Key	Notes
`id`	int	PK	
`settlement_line_id`	int	FK	
`returned_amount`	decimal		
`bank_fee`	decimal		
`reason`	string		
`returned_at`	datetime		

CONFIRMATION

Field	Type	Key	Notes
`id`	int	PK	
`settlement_line_id`	int	FK	
`amount_received`	decimal		
`slip_ref`	string		
`result`	string		full / short
`confirmed_by`	int	FK	
`confirmed_at`	datetime		

SHORTFALL

Field	Type	Key	Notes
`id`	int	PK	
`receiver_id`	int	FK	
`confirmation_id`	int	FK	
`amount`	decimal		
`state`	string		open / topped_up
`topped_up_settlement_id`	int	FK	

BANK_FEE_EVENT

Field	Type	Key	Notes
`id`	int	PK	
`settlement_id`	int	FK	
`expected_amount`	decimal		
`actual_amount`	decimal		
`treatment`	string		waived / carry_forward / absorbed
`state`	string		open / applied
`applied_settlement_id`	int	FK	

FEE_DECISION

Field	Type	Key	Notes
`id`	int	PK	
`subject_type`	string		Deal / Settlement / BalanceConversion
`subject_id`	int		
`fee_kind`	string		markup / network / bank
`decision`	string		apply / waive / override
`pct`	decimal		
`recommended_by`	int	FK	Finance — reviews and recommends before Management approves a covered bank fee or a markup above the cap
`approved_by`	int	FK	Management for a covered bank fee or a markup above max_markup_pct; Finance for the other one-off decisions
`reason`	string		

MARKUP_SHARE_ACCRUAL

Field	Type	Key	Notes
`id`	int	PK	
`markup_share_rule_id`	int	FK	
`deal_id`	int	FK	
`amount`	decimal		
`state`	string		accrued / reconciled / settled / paid)
`paid_at`	datetime		
`direction`	string		payable (NPL pays the party) / receivable (the party holds the funds and owes NPL its remainder)
`period`	string		YYYY-MM; receivables reconcile monthly like rebates
`received_amount`	decimal		on receivables

PARTNER_REBATE_ACCRUAL

Field	Type	Key	Notes
`id`	int	PK	
`partner_rebate_rule_id`	int	FK	
`deal_id`	int	FK	
`expected_amount`	decimal		
`received_amount`	decimal		
`period`	string		YYYY-MM
`state`	string		expected / reconciled / received
`statement_ref`	string		the partner statement line it reconciled against

LOSS_EVENT

Field	Type	Key	Notes
`id`	int	PK	
`side`	string		collection / settlement
`collection_id`	int	FK	when side = collection
`settlement_line_id`	int	FK	when side = settlement
`amount`	decimal		≤ the verified or released amount

Field	Type	Key	Notes
`reason`	string		account_frozen / cash_short / wrong_chain / other
`occurred_at`	datetime		
`sender_share`	decimal		
`npl_share`	decimal		
`partner_share`	decimal		
`client_share`	decimal		shares sum to amount
`state`	string		open / approved / posted
`approved_by`	int	FK	Management
`pool_funded`	decimal		the part of the make-good taken from NPL's pool at the partner
`wallet_funded`	decimal		the part transferred from NPL's own wallet to the partner; what neither covers is the unfunded part, posted as MAKEGOOD / LOSS_PAYABLE until NPL's funds land
`funding_reference`	string		the partner's credit note or the own-wallet transfer behind the make-good

LOSS_RECOVERY

Field	Type	Key	Notes
`id`	int	PK	
`loss_event_id`	int	FK	
`party`	string		sender / partner / client
`amount`	decimal		≤ the matching share on the LOSS_EVENT
`state`	string		agreed / received — a client share is received at once (it is a reversal of the make-good, never a payment). Each state change posts one LEDGER_TRANSACTION naming this row: agreed raises LOSS_RECEIVABLE against EXP_LOSS (or, for the client, reverses part of the make-good), received clears the receivable into POOL or WALLET
`received_at`	datetime		
`reference`	string		the partner's credit, the sender's transfer, the client's agreement
`approved_by`	int	FK	Management (approve_loss_split)

B.5 Ledger & controls

LEDGER_TRANSACTION

Field	Type	Key	Notes
`id`	int	PK	
`source_type`	string		Collection / Conversion / BalanceConversion / Settlement / SettlementReturn / Confirmation / BankFeeEvent / Reroute / MarkupShare / Rebate / LossEvent / LossRecovery / Invoice / EarningsOffset / Adjustment
`source_id`	int		
`description`	string		
`posted_at`	datetime		
`posted_by`	int	FK	
`reversal_of_id`	int	FK	

LEDGER_POSTING

Field	Type	Key	Notes
`id`	int	PK	

Field	Type	Key	Notes
`ledger_transaction_id`	int	FK	
`ledger_account_id`	int	FK	
`side`	string		debit / credit
`amount`	decimal		
`currency`	string		
`ownership_tag`	string		client / company
`cost_component`	string		principal / earnings / rebate / partner_cost / bank_fee / network_fee / variance / share / rounding / fx_timing / cutoff_timing / loss / resale
`receiver_group_id`	int	FK	nullable; set on every CL.PROJECT.<CCY>.PAYABLE line: the leg's group at collection and conversion, the paying leg's group on a release, return or short confirmation (SETTLEMENT_LINE.deal_id), the served group on a balance conversion, the deal's group on a loss share or an earnings offset — so GROUP_ENTITLEMENT is a sum over postings

LEDGER_ACCOUNT

Field	Type	Key	Notes
`id`	int	PK	
`code`	string		OWNER.HOLDER.CCY.PURPOSE
`ownership`	string		client / company
`holder_type`	string		PartnerConfig (incl. the own desk) / OwnWallet / Receiver (also the vendor of VENDOR_PAYABLE) / Sender (also LOSS_RECEIVABLE, RESALE_RECEIVABLE) / MarkupShareParty / Project / None
`holder_id`	int		
`currency`	string		
`purpose`	string		COLLECTED / HELD / DUE / PAYABLE / SHORTFALL / CREDIT / MAKEGOOD / POOL / WALLET / EARN_GROSS / EARN_REBATE / RESELL_REVENUE / RESELL_COST / EXP_PARTNER / EXP_BANKFEE / EXP_NETWORK / EXP_ROUNDING / EXP_LOSS / VAR_CONVERSION / VAR_CUTOFF / VAR_RATE_HONOUR / VAR_FX_TIMING / SHARE_PAYABLE / SHARE_RECEIVABLE / REBATE_RECEIVABLE / LOSS_RECEIVABLE / LOSS_PAYABLE / VENDOR_PAYABLE / RESALE_RECEIVABLE / REMIT_CLAIM / FX_RESULT
`kind`	string		asset / liability / income / expense; FX_RESULT is income or expense by sign, per currency

USER

Field	Type	Key	Notes
`id`	int	PK	
`email`	string		Google Workspace
`role_id`	int	FK	ROLE free-text role)
`status`	string		

APPROVAL

Field	Type	Key	Notes
`id`	int	PK	
`subject_type`	string		
`subject_id`	int		
`action`	string		

Field	Type	Key	Notes
`requested_by`	int	FK	
`approved_by`	int	FK	never equal to requested_by
`escalated_to`	int	FK	
`state`	string		
`reason`	string		

AUDIT_LOG

Field	Type	Key	Notes
`id`	int	PK	
`record_type`	string		
`record_id`	int		
`field`	string		
`old_value`	string		
`new_value`	string		
`actor_id`	int	FK	
`at`	datetime		

RECORD_LOCK

Field	Type	Key	Notes
`id`	int	PK	
`record_type`	string		
`record_id`	int		
`locked_by`	int	FK	
`locked_at`	datetime		

ROLE

Field	Type	Key	Notes
`id`	int	PK	
`code`	string		operations / finance / management
`name`	string		
`sees_economics`	bool		false for operations: no derived-economics fields on any screen, export or message
`status`	string		

ROLE_RIGHT

Field	Type	Key	Notes
`id`	int	PK	
`role_id`	int	FK	
`action`	string		a controlled action: price_quote, extend_quote_validity, cross_group_payout, cover_bank_fee, enable_own_desk, approve_loss_split, override_screening_reject, earnings_offset, …
`may_initiate`	bool		
`may_approve`	bool		
`escalates_to_role_id`	int	FK	nullable; the role added on escalation